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A Systematic Review of the Current State of Transfer Learning accelerated CNN-Based Plant Leaf Disease Classification

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ABSTRACT Crops and their produce are vital to the livelihood of humans everywhere. World food security heavily relies on them, but still, even today, hundreds of millions of people world-wide are suffering from hunger. This is why it is essential to ensure that losses to the agricultural yield are kept at a minimum. Plant diseases, however, cause massive losses to the possible yield every year, rendering large amounts of the planted crops useless. And, if these diseases are not identified early enough, they will further infect more plants and therefore destroy even more yield. This is why plant diseases need to be recognized as fast as possible. Many diseases can be detected via the symptoms present on the plants leaves. As such, traditionally, staff samples and checks plants for their health in field manually. To speed up and increase accuracy, deep learning methods have been proposed to classify plant leaf images by their diseases. To train such models, however, one needs sufficiently large datasets. Such datasets are a gap in the field, with many datasets either being too small, not applicable (wrong plants or diseases), private, or taken under conditions that do not apply to the task at hand. One way to lower the need for more data and to overcome lesser data availability is transfer-learning, which utilizes unrelated rich, big, and available data to train feature extractors, which can then be re-trained to identify plant-leaves, based on the knowledge extracted from the prior task. In this work we will review and analyze a total of 84 CNN based transfer-learning papers from 2022 to 2025 out of 118 considered using PRISMA and discuss the insights gathered from them. This paper presents statistics on model, dataset, hyperparameter, pre-processing, augmentation, and metrics usage. Also, different methods for improving TL CNN models and training are listed. Additionally, performance comparisons of different prominent models for the field of plant leaf disease classification, as well as performance comparisons of different dataset types will be provided.

INDEX TERMS Artificial Intelligence, CNN, Crop Disease Detection, Deep Learning, Image Classification, Plant Disease Detection

I. INTRODUCTION

Even today world hunger is still a huge problem that is affecting hundreds of millions of people. Recent reports show that the number of people affected by undernourishment has risen by 1.6% compared to just 5 years ago [1], a concerning trend. Africa is most heavily affected by undernourishment with over 20% of the population suffering from it [1].

Crops are essential to assure food security to as many people as possible all across the world. Since 2000 crop production has increased by over 50% to a point where plant based products are responsible for over 80% of the calories consumed worldwide and for around 60% of the worldwide protein supply [2]. On top of that, outlooks predict that the demand for food obtained through agriculture will only rise

[3] in upcoming years. These statistics prove the high significance and importance that agricultural production and crops have on the food supply and showcase that it is pivotal that their production is secured and assured.

Crop diseases and pests are, however, a huge danger to the security of these agricultural crop yields. Almost 40% of the annual crop production are lost to such pests and diseases [3]. All these crops being lost prematurely is a large waste of potential food supply and reducing the amount lost as far as possible should be a goal of high priority.

Detecting plant diseases needs to be done as soon as possible, to minimize the negative effect they will have on the yield [4]. Traditionally, and commonly, field scouting is used to locate and identify plant diseases in the fields [5]. This process requires trained personnel to go out into the fields and manually look at plants and check them for symptoms that could indicate disease infections. This process is rather laborious and tedious and needs to be repeated multiple times per field during the life-cycle of the plants [6].

One way of speeding up the process, while also assuring high accuracy has been introduced with the application of deep learning (DL) artificial intelligence (AI) models that analyze the plants for diseases based on images taken of the plants leaves [7]–[12]. This is made possible by recent advances in computer vision and image classification. Convolutional neural networks (CNN) are to thank for that, with their efficient and effective way of automatically training feature detectors. New models that are faster, better performing, more lightweight and more optimized have only further improved the capabilities that such AI models can have in the field of automated plant leaf disease identification. Models such as MobileNet [13], EfficientNet [14], and DenseNet [15] are some of the models that have recently found successful application in plant leaf disease classification research.

One important prerequisite to training high-performing models is adequate datasets. These datasets need to be large enough, diverse enough, and high quality enough so that the model can learn proper representations of the plants and their diseases and then generalize that knowledge to before unseen images. Prior research, however, points datasets out as an area of concern in plant leaf disease classification [8], [9], since not enough high quality datasets are openly available. Additionally, some of the most popular datasets (e.g. PlantVillage [16]) are taken in lab conditions, which makes them less reliable in real field conditions [17]–[20].

One way to overcome smaller data is transfer learning (TL) (see Figure 1). Transfer learning utilizes pre-trained weights, based on large-scale image datasets unrelated to the final tasks. This is often done by training CNN models on the ImageNet dataset [21] first, a dataset with 1,000 classes and over 1 million images. Through this training, the models will learn general feature maps, both high and low level, that can then either be directly applied to a new task, or which can be adapted and optimized to better work with the new data. As such, the dataset needed for training can be smaller, since all the prior knowledge obtained already provides much more

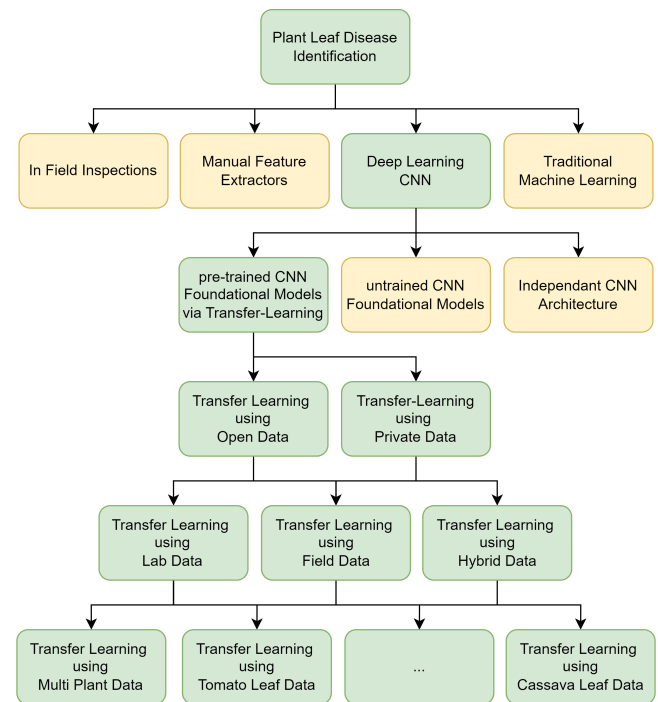


FIGURE 1. Conceptual Diagram of Methods for Plant Leaf Disease Classification

robust features than a newly and randomly initialized model would.

In previous literature review papers publish in the field of plant leaf disease classification using ML and DL methods Nagaraju et al. [7] review deep learning methods to detect plant leaf diseases in their paper. It puts focus on data acquisition and also offers insights into the usage of different plant species, pointing out that vegetables and fruits are the most commonly studied plants. The paper however puts little focus on models, hyperparameters, and general datasets and also only considers papers published between 2015 and 2019. Li et al. [8] review papers focused on detecting and classifying plant diseases. The dataset puts focus on working with limited data and as such focuses on few-shot learning, as well as TL. The paper does, however, not go into much detail on TL and also does not offer insights into model usages, dataset usages, or other parameters related to the training of models. Lu et al. [9] review CNN models applied to the field of plant leaf disease classification specifically. The paper lists advantages of using CNN in the field and gives pointers to the construction of new models (such as layer depth). The authors also point out the heavy usage of the PlantVillage dataset and the disadvantages that come from using lab only datasets. The paper does, however, not share model or dataset usage statistics, or any other best practices for training. Sarkar et al. [10] review machine learning and deep learning based leaf disease detection methods in their paper. They put heavy focus on different plant types and diseases that these plants can be exposed to in their work, explaining plant-disease pairs and their symptoms in great detail for numerous such com-

binations. The model also tracks research origins, but does not track detailed model usages, augmentation techniques or dataset usages. Demilie et al. [11] review techniques for plant leaf disease detection and classification. The author points out that DL methods outperform traditional ML models in the field, but also notes that hybrid models (e.g. CNN feature extractor with SVM classifier) also tend to perform well. The paper also mentions that at the time, out of the reviewed papers, tomato plants were the most studied species. The paper does not put much emphasis on CNN-specific models or TL, and does not analyze model usage, the most used pre-processing techniques, or other best training practices.

Based on these previous literature review papers in the field and the information they provide, research gaps in terms of model usage statistics and performance metrics and best practices, dataset usage numbers, and other information on best practices for model training (such as hyperparameters, augmentation, pre-processing, epochs, etc.) have been identified. Additionally, none of the mentioned papers offer a clear focus on CNN methods, TL, and classification. As such, in this paper we will focus on providing information on model and dataset usage statistics (as well as listing and explaining these models and datasets with their key values), offer insights into hyperparameters and other training related issues, and focus specifically on TL CNN for classification, thus allowing for more depth in this specific field and for better comparability among the studies reviewed.

The resulting research questions that will be asked and answered in this paper are the following:

- **Q1:** Which pre-trained models are currently most often used and best suited in terms of performance?
- **Q2:** Which dataset types are currently most often used and best suited in terms of performance?
- **Q3:** Which methods can be used to improve pre-trained models?
- **Q4:** Which hyperparameters are currently most often used and best suited?
- **Q5:** Which metrics are currently most often used and best suited?
- **Q6:** Which pre-processing and augmentation techniques are currently most often used and best suited?
- **Q7:** Which Transfer-Learning strategies are currently most often used and best suited?

A. PAPER SELECTION PROCESS

In this work, we will review a total of 84 papers that were selected from a pool of 118 initial papers, introduce each one of them analyze common and popular trends, identify best practices and point out common methods and datasets.

The papers were obtained from the Web of Science database by using the search-term '("Plant Disease" OR "Leaf Disease") AND "CNN" AND "classification" AND "transfer-learning"' and by filtering the search to 2022, 2023, 2024, and 2025 to catch the most recent trends in the field and because the field started gaining much more research attention in 2022 (see Figure 2 - The drop-off in 2025 is simply because it is

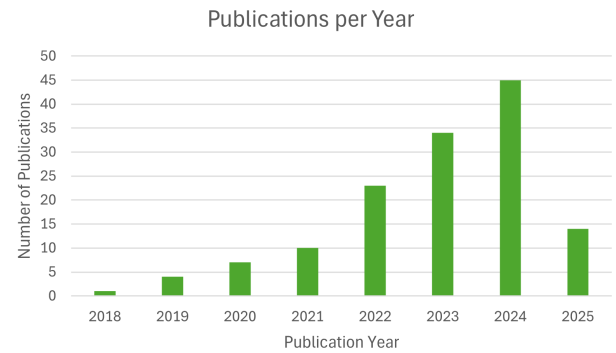


FIGURE 2. Papers published per year. Data obtained on Web of Science via the search-term '("Plant Disease" OR "Leaf Disease") AND "CNN" AND "classification" AND "transfer-learning"'

only midyear 2025 at the time of writing), with many more papers being published in said span than they were before 2022 (this can probably be somewhat attributed to the focus of transfer-learning, which itself is becoming more popular in recent years). Additionally, reviewing more recent years will produce findings that are more recent and more up-to-date with current trends and best practices. This search yielded 118 original papers. Paper were then filtered based on the criteria listen below, following the PRISMA approach:

- Only papers that were published in English
- Only papers that were accessible (published on major platforms or in open access)
- Only papers that made use of Transfer-Learning
- Only papers that utilized CNN
- Only papers that used plant leaf images as data
- Only papers that work with classification models
- Only experimental research papers (No literature reviews)

Only english papers were considered to make sure that we could properly understand the manuscripts without having to rely on translation apps and the few papers we were not able to access had to be excluded since one can not summarize a paper only based on the abstract. Only reviewing papers that include TL, CNN, plant leaf image data and classification models is based on the scope of this paper, as this review is centered around TL for CNN based plant leaf disease classification. Lastly, review papers were not included in the review section itself, as they do not introduce any new experimental results.

Out of these 118 original papers we selected the papers for review based on the PRISMA method (see Figure 3 and listing above), which resulted in 84 papers that are being discussed in this work.

Due to the the diversity of models, datasets, metrics, hyperparameter, and training configurations, these 84 papers will be analyzed and contextualized through the application of quantitative synthesis approaches, through the aggregation of model, dataset, augmentation, and hyperparameter usage

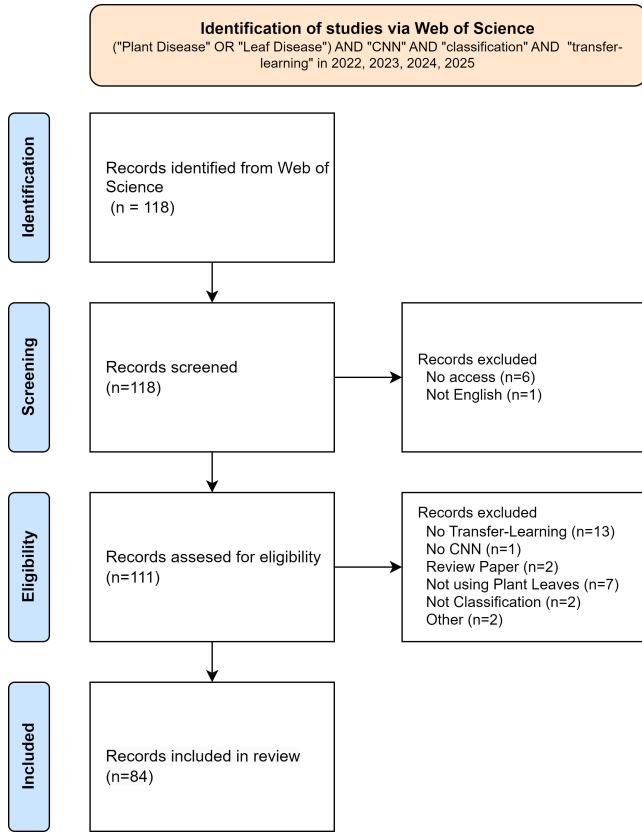


FIGURE 3. PRISMA Diagram showcasing the paper selection process. Papers were originally obtained from the Web of Science with the search-term ("Plant Disease" OR "Leaf Disease") AND "CNN" AND "classification" AND "transfer-learning" for 2022, 2023, 2024, and 2025.

statistics and through the comparison of relative performance using a heuristic quartile based ranking method.

As such this work's contributions lie in:

- An extensive and systematic review of current literature (84 papers) in the field of TL plant leaf disease classification
- Analysis of current trends in TL for agricultural applications, offering insights into the evolving research landscape
- Identifying and highlighting the most common methodologies and best practices, offering a detailed overview of effective strategies in the domain
- Compiling and examining the most frequently used CNN architectures and datasets with various disease classes, facilitating a clearer understanding of model selection and dataset preferences in agricultural AI research

a: Screening and Quality Assessment

To ensure the relevance and quality of the included studies, a two-stage manual screening process was conducted following PRISMA guidelines. The initial screening involved evaluating article titles and abstracts to exclude irrelevant studies. Two reviewers independently screened records retrieved from

Web of Science, which resulted in 118 potentially relevant articles. In the second step, three reviewers conducted a full-text evaluation based on predefined inclusion criteria, ultimately selecting 84 articles for final analysis. The complete literature screening process is illustrated in Figure 3.

To minimize bias and maintain methodological rigor, we applied two levels of assessment: Research Content Criteria and Literature Quality Criteria, as detailed in Table 1. Only peer-reviewed studies focusing on CNN and transfer learning based plant disease detection or diagnosis were included. Articles merely mentioning the topic without substantial analysis were excluded. Furthermore, all included studies were required to be full-text accessible, contain essential metadata, and meet peer review standards.

b: Validity and Reliability

To ensure the validity of our systematic review, multiple dimensions were addressed:

Internal Validity: A structured and transparent selection process was implemented, including predefined inclusion criteria (see Table 1), manual screening by multiple reviewers, and PRISMA-compliant documentation of all steps.

Construct Validity: Studies were categorized based on clearly defined criteria relevant to CNN and transfer learning for plant disease detection, ensuring accurate reflection of core research themes.

External Validity: The review draws on diverse datasets, crops, and geographic regions, supporting the generalizability of findings to broader agricultural contexts.

Conclusion Validity: Findings were based on aggregated evidence from peer-reviewed studies, with consistent patterns emerging across datasets and models. Interpretations were cross-verified among co-authors to avoid bias.

B. MANUSCRIPT STRUCTURE

The rest of the paper is structure as follows: Section II will introduce the field of deep learning based plant leaf disease classification using transfer learning accelerated CNN and explains the basics behind it. Section III includes the actual review, where each paper will be summarized, after which Section IV will analyze and contextualize all the information gathered from said papers. Section V will outline possible future directions for the field before Section VI concludes this work.

II. BACKGROUND

As mentioned earlier, a key component required to train well performing models are sufficient datasets.

A. DATASETS

As pointed out in prior work, datasets are still an area of need in the field [8], [9]. This is because a large number of datasets used in research are kept private and closed source, meaning they are not open to be used by others. Insufficient data, be it not enough images, images taken in conditions not applicable to the field, data with heavy bias, etc. can lead

TABLE 1. Inclusion and Quality Assessment Criteria

Type	Inclusion Criteria
Research Content Criteria	1. The article must focus on CNN transfer learning based plant disease detection or diagnosis as the central research topic. 2. Articles that mention the topic only briefly or in background context were excluded.
Literature Quality Criteria	1. Only peer-reviewed articles were included (Original Research, Review). 2. Full-text availability from authoritative databases. 3. Articles must include complete metadata (abstract, author info, references). 4. Editorials, preprints, and opinion papers were excluded unless methodologically relevant.

to a wide range of issues that could either become evident during training or during testing/application in field. Limited data could lead to overfit models, models that do not manage to learn real robust representations of the actual plant leaf diseases and are therefore essentially useless in testing or production. Techniques such as few-shot learning [22], [23] can be utilized to potentially limit the negative effects of this. In terms of heavy imbalance in datasets, which can lead to biases in the model (meaning the model learns to simply predict the most common classes instead of learning true features), techniques such as data augmentation, synthetic data generation, or class weighting can be applied in an effort to limit the biasing [22]. As for images that are taken in insufficient conditions (training data only contains lab images, but application will be in field), or if the training data is incomplete in the sense that it does not contain all classes required, one could pre-train the model on that insufficient data first, before fine-tuning it to the target dataset (using few-shot learning for example, if that target dataset is small) which can help bridge the gap in domains in domain shift [22]. Across the openly available datasets that are free to use by the research community, we can mainly divide them into 2 subcategories, these are datasets taken under lab conditions and images taken in field. Additionally, hybrid datasets that combine the 2 also exists and see use. A list of datasets that saw use in the papers reviewed in this work can be seen in Table 2.

1) Lab-Based Image Datasets

Images taken under lab conditions are images where the leaves are taken in front of a perfect background, often by taking single leaves from the field (plucking them from the plant) and putting them on a single colored table, where the images are taken (see Figure 4). These images are rather simple for DL models to learn. This is because the background poses no challenge, being so distinctly different from the leave itself, while also being without any sort of noise or distracting features. This leads to models usually achieving very high scores on datasets taken in lab (as can be seen in later sections), often making it hard to really benchmark their performance due to these high scores. Models trained on lab data is also said to underperform when used in field conditions [17]–[20], [22].

2) Field-Collected Image Datasets

Field datasets, as the name suggests, are datasets that were collected in field under real conditions. This means that these datasets are much more complex, with the background of the leaves containing other plants and their leaves, soil, branches, or other details that are not the main focus of the image (see Figure 5). As such, the model needs to be able to distinguish the leaf in the foreground from the rest of the image and then only consider that for classification, making the task much tougher. Additionally, the images are also taken in less constant and ideal conditions in terms of angles, lighting, etc. This makes learning more difficult and benchmarking easier because not almost all model reach 90%+ almost by default, as they often do in most lab datasets [45]–[47]. And, these models are also trained to be actually used in field.

3) Hybrid-Condition Image Datasets

Hybrid datasets are datasets that contain images taken under both conditions (lab and field). An example of such a dataset can be seen in Figure 6, where sugarcane images were collected in field and in lab. As such, a resulting dataset, a dataset with images taken in both conditions, is exposed to more diverse data (data that was taken in field and in lab), which can lead to models that generalize better and perform better with new data in various conditions, since it was already exposed to various conditions during training, leading to more robust features [18]–[20].

B. MODELS

CNN are some of the most powerful and best suited models to handle image input data for classification tasks.

1) CNN

Unlike fully connected multi layer perceptrons (MLP), CNN are specifically designed to handle image data and are, as a result of that, much more efficient in image classification tasks. To achieve that, they use automatic feature extractors. Unlike before CNNs, where feature extractors were coded by hand, in CNNs these feature extractors get learned automatically through back-propagation. Kernels, which are usually squares made up of learnable parameters, move over the image from top left to top right, across all layers (see Figure 7). These kernels then get multiplied with the pixel values stored in the

TABLE 2. List of the most commonly used open datasets.

Dataset	Num. Images	Num. Classes	Plant	Conditions
PlantVillage [16]	54,304	38	Multi	Lab
Rice Leaf Disease Image Samples (RLD) [24]	5,932	4	Rice	Hybrid
Potato Leaf Disease Dataset (PLDD) [25]	1,500	3	Potato	Lab
Rice Leaf 5 diseases [26]	2,710	6	Rice	Lab
PlantDoc [17]	2,577	28	Multi	Hybrid
CD&S Dataset [27]	1,571	3	Corn	Field
BRACOL [28]	1,747	5	Coffee	Lab
DiaMOS [29]	3,006	4	Pear	Field
iCassava [30]	21,397	5	Cassava	Field
Rice Disease Dataset DMD [31]	3,424	5	Rice	Hybrid
Rice Leaf Diseases Dataset [32]	120	3	Rice	Lab
FGVC7 [33]	1,821	4	Apple	Field
Wheat Disease Detection [34]	3,679	3	Wheat	Field
JMuBEN [35]	22,591	3	Coffee	Lab
Chili Plant Disease [36]	500	5	Chili	Field
Paddy Doctor [37]	10,407	10	Rice	Field
Banana Leaf Disease Images [38]	1,288	3	Banana	Field
Blackgram Plant Leaf Disease Dataset [39]	1,000	3	Black Gram	Field
MangoLeafBD Dataset [40]	4,000	8	Mango	Lab
Tomato leaf disease detection [41]	11,000	10	Tomato	Lab
Date Palm data [42]	2,631	3	Date	Lab
Olive Leaf Disease [43]	954	2	Olive	Lab
Enset Dataset [44]	999	2	Enset	Field



FIGURE 4. Examples of raspberry plant leaf images taken in Lab. Images obtained from PlantVillage Dataset [16].



FIGURE 5. Examples of pear plant leaf images taken in Field. Images obtained from DiaMOS Dataset [29].

image at the current location and then they all get added up to form the final new value post feature extraction. This kernel moves across the image from the top left to the bottom right and keeps applying the kernel to the selection every time. The resulting new matrix of values is the feature map, the image adjusted via the kernels, which will be passed on to deeper layers. These can either be more convolutional layers or fully connected layers in the end to classify. Deeper convolutional layers tend to learn more fine grained details, while earlier layers often learn more general and simple features. Through these kernels then number of parameters is kept low, while also considering spatial information through them. Each model comes with different architectures and sizes (see

Table 3) and can be trained with hyperparameters depending on the task at hand and for as many epochs as the dataset requires.

Some of the most popular and most used CNN models, that are also among the most used models in this review (see Table 12 and Figure 8) will be introduced and explained in the paragraphs below and a list of models can be seen in Table 3.

α : LeNet

LeNet [49] is a CNN architecture that has been introduced way before the big deep learning boom in the 2010s. It was conceptualized in 1998 and is rather simple in its structure. It is made up off 2 blocks of convolutional layers followed by



FIGURE 6. Examples of sugarcane leaf images taken in Hybrid conditions. Images obtained from Sugarcane leaf dataset [48].

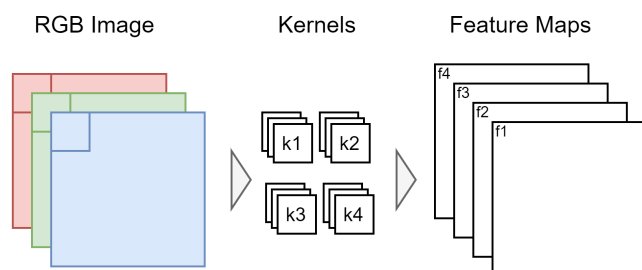


FIGURE 7. Visualization of how a CNN and its kernels work to generate feature maps.

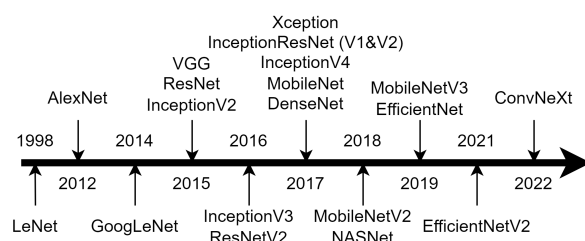


FIGURE 8. A timeline of some of the most popular CNN models.

pooling layers, before the dense layer classifier.

b: AlexNet

AlexNet [50] is a model that is not too far of the LeNet mentioned above. It too features only convolutions layers and pooling layers. The first 4 layers are also 2 blocks of convolutions plus pooling, followed by 3 convolutional layers after which there is another pooling layer. Next is the classifier.

c: Inception

The Inception [51] architecture, which was first known as GoogLeNet, introduced factorized convolutions. These factorized convolutions work differently from the regular square kernels. Regular square kernels would be split up into two single row-like kernels. A 5x5 square kernel for example could be split up into a 3x1 and a 1x3 kernel, which allows Inception the reduce the number of parameters needed. It also introduced multi-scale-processing, which means that differently sized kernels were used inside the network. Lastly, batch-normalization was used to normalize the output of layers.

d: VGG

VGG [52] is another network that uses only regular convolutions and pooling layers, just like LeNet and AlexNet. The big difference between the VGG architecture and the previously mentioned is that VGG is much deeper than AlexNet and LeNet, stacking many more convolution and pooling blocks.

e: ResNet

The ResNet [53] family of models uses residual connections within its architecture to allow early features to residually impact later layers. To achieve this, the layer's input will be added to the layer's output. To make sure that additions are possible, padding is added to the convolution layers to ensure that the output size is identical to the size of the input.

f: MobileNet

MobileNet [13] is a architecture that aims to be efficient and that aims to require less computational cost than other networks. To do this, a few new techniques were introduced. For one, MobileNets use depthwise seperable convolutions. These combine depthwise and pointwise convolutions. Where a normal kernel is as deep as the input image, a depthwise kernel is only looking at a single input channel (a regular kernel operates on all). Pointwise kernels on the other hand are only 1x1 pixels in width and height, but span all channels in depth. This combination allows the kernel to still grasp spatial information in all 3 dimensions of the data, while being less parameter costly.

g: DenseNet

The DenseNet [15] architecture is an architecture that, not unlike ResNet, focuses on residual connections. However, here there are no additions. In DenseNet the input of the layers all get stacked. This means that the input keeps getting deeper because more and more channels keep being appended. To do this, here too the padding needs to be enabled and the size of the image can not be changed within a dense block, only in-between different different dense blocks, so that the stacking remains possible.

h: NASNet

NASNet [54] is a CNN that was built using optimizing approaches like reinforcement learning to find the best performing NASNet structure, which was made up of different blocks and cells. These blocks had different types (regular

convolutions, separable-convolutions, max pooling, average pooling, etc.).

i: EfficientNet

The EfficientNet [14] architecture is one that utilizes different widths and depths that allow researchers to pick one that best suits their needs. It utilizes the inverted bottleneck method, which uses a number of pointwise convolutions before the depthwise-separable convolution layers to increase the number of channels of the input. It also uses stochastic depth, which drops complete layers at random to increase robustness, unlike regular dropout that just drops single neurons (which are also used).

j: ConvNeXt

ConvNeXt [55] is designed based on the idea of ResNet, in an effort to build a CNN that makes use of the advantage of vision transformers, while maintaining the advantages that CNN come with. To do that it utilizes larger than normal kernel sizes and layer-normalization (instead of the more common batch-normalization). Techniques like the reverse bottleneck, depthwise convolutions and residual connections are also used, and ConvNeXt also uses GELU activations.

2) Transfer-Learning

Transfer-Learning (TL) is a method for CNN and DL that comes with a certain set of benefits. With TL, one can train their models faster, with less data, and potentially also train them to be stronger and more robust. This is due to the fact that TL uses weights, and therefore knowledge, obtained from previous tasks and transfers that knowledge to the new domain specific task. Commonly, this is done as follows: the CNN architecture is trained on the ImageNet dataset [21], a dataset of over 1.2 million images across 1,000 classes. This initial training exposes the model to many different images and image types, and based on those the model can learn many different feature extractors that are diverse and robust. These feature extracting layers and their weights can then be downloaded, copied, and applied to new tasks (see Figure 9). The classifier top, the last layers in the DL model (often fully connected layers), need to be replaced and adjusted to fit the new tasks (they need to match the number of classes), but the CNN feature extractor can remain unchanged. One can also either freeze (prevent their weights from updating) all or some of the layers of the CNN to speed up training and to make training more lightweight, or allow all or some of the to be retrained based on their preferences. Overall, these techniques compensate for lack of domain-specific data, as many relevant features are already learned, lower the chances of overfitting, by having been exposed to very diverse data previously, and can speed up training, by nature of already having been trained in the past. On top of that, training large models with TL can be less hardware intense, since one can freeze layers, meaning they do not need to be re-trained.

TABLE 3. List of the most commonly used models (see Figure 10) in TL plant leaf disease detection, their reference and some key values (without the classifier top). Memory is listed in MB, and M in parameters stands for million.

Model	Total Params	Total (MB)	Mem.
AlexNet [50]	3.75M	14.29	
DarkNet53 [56]	40.62M	154.96	
DenseNet121 [15]	7.04M	26.85	
DenseNet169 [15]	12.64M	48.23	
DenseNet201 [15]	18.32M	69.89	
EfficientNetB0 [14]	4.05M	15.45	
EfficientNetB1 [14]	6.58M	25.08	
EfficientNetB2 [14]	7.77M	29.63	
EfficientNetB3 [14]	10.78M	41.14	
EfficientNetB5 [14]	28.51M	108.77	
EfficientNetB7 [14]	64.1M	244.51	
GoogleNet [51]	2.74M	10.47	
InceptionV3 [57]	21.80M	83.17	
InceptionResNetV2 [58]	54.34M	207.28	
LeNet [49]	0.05M	0.19	
MobileNet [13]	3.23M	12.32	
MobileNetV2 [59]	2.26M	8.61	
MobileNetV3-Small [60]	0.94M	3.58	
NASNetLarge [54]	84.92M	323.93	
NASNetMobile [54]	4.27M	16.29	
ResNet101 [53]	42.66M	162.73	
ResNet152 [53]	58.37M	222.67	
ResNet152V2 [61]	58.33M	222.52	
ResNet18 [53]	11.19M	42.67	
ResNet34 [53]	21.3M	81.26	
ResNet50 [53]	23.59M	89.98	
ResNet50V2 [61]	23.59M	89.98	
SqueezeNet [62]	0.74M	2.81	
ShuffleNet [63]	0.96M	3.67	
VGG16 [52]	14.71M	56.13	
VGG19 [52]	20.02M	76.39	
Xception [64]	20.86M	79.58	

a: Fine-Tuning

TL tasks can be carried out in a variety of different configurations in terms of how TL is applied to the baseline pre-trained model by deciding how much of said model to freeze. Here freezing refers to making layers "un-trainable", meaning they will be considered during forward runs through the model during "inference" but not during backwards-propagation (training). As such, frozen layers always keep their original pre-trained weights, which can offer stability against overfitting and more light-weight training in terms of hardware requirements, but at the cost of potentially better tuned and better performing models on the target domain data. Fine-tuning, that is un-freezing layers to let the adjust to the new data, can improve performance, but can lead to overfitting. Commonly used approaches are completely freezing all CNN layers and only training the classifier top (not utilizing fine-tuning), freezing only a certain amount of layers (often earlier ones) and unfreezing and fine-tuning the rest, or not freezing any layers and fine-tuning the entire model. Additionally, one could do this in episodic fashion, where, for example, at first all layers are frozen but then all layers are unfrozen for fine-

tuning. Applicability often depends on the dataset size and domain similarity between the pre-trained domain and the target domain.

b: Potential Drawbacks

But TL does also come with some potential drawbacks. These include domain mismatches and shifts, pre-trained biases, and possible constraints in model architecture. Domain shift issues can arise when the pre-train model weights were trained on a dataset that is very different from the domain specific dataset that one wants to train the model on, where the pre-trained model does not benefit from pre-training, but rather suffers from it in certain circumstances. In cases like this, domain adaptation techniques (e.g. the right fine-tuning strategy, data augmentation to bridge domain gaps, choosing the right pre-trained weight origins) could be useful to overcome these issues. In this field ImageNet pre-trained models tend to work well in TL settings, but using a more domain-related dataset and domain adaptation methods could further improve results and domain adaptation robustness and lessen the need for domain adaptation. Pre-trained bias can occur when the model is pre-trained on bad data and imbalanced data. In such a case the pre-trained weights would contain heavy biases and be useless, if not worse, for TL. ImageNet, as a dataset, is well balanced enough as to where this should not be a problem, but when using a different dataset for pre-training, one should make sure to pay attention to this. Restraints in architecture refer to the fact that TL is often carried out by downloading pre-trained weights for already existing model architectures rather than building a new model and then training that model on ImageNet before applying TL. As such, one is mostly limited to using said models, and modifications to that model are also mostly limited to adding layers and mechanisms to the end. This could be overcome by pre-training the model after the architecture is finalized. But generally speaking, the available models with weights already offer strong architectures with good performance.

Overall it can be said that TL comes with many advantages, but not without drawbacks. It should however be pointed out that, if done correctly, in most scenarios the advantages with heavily outweigh the disadvantages, and that most drawbacks can be mostly overcome if one applies TL carefully and thoughtfully.

3) Transfer-Learning for Plant Leaf Disease Detection

These TL methods explained above have too seen previous usage in the field of plant leaf disease detection, where different pre-trained models have been applied to a wide range of different datasets, containing a multitude of different plant species and disease types. Datasets, be it field, hybrid, or lab, have been utilized to tackle this task and the pre-trained models (most often trained on ImageNet) show promise and good results in a lot of different setups (different models, datasets, hyperparameters, plant, diseases, etc.). More information on the specific usage of TL in plant leaf disease classification can be found in the following sections.

III. LITERATURE REVIEW

In this section we will review the individual papers (all 84). The papers will be categorized by dataset. First grouped by open and private datasets before being further divided into field, lab, and hybrid dataests. Within these categories, the papers will be ordered by publication date.

A. PAPERS USING OPEN IMAGE DATASETS

First, we look at papers that use open datasets, which are publicly available online. In the following sections, we discuss each category consecutively.

1) Papers using Open Lab-Based Image Datasets

a: Papers using Open Lab-Based Tomato Leaf Image Datasets

Hammou et al. [65] train both a pre-trained DenseNet169 [15] model and InceptionV3 [57] model on all tomato class images contained in PlantVillage [16], [66]. This results in a dataset of 10 classes (9 disease and 1 healthy) with over 18,000 images across all of those classes. Images were first split into 20% for testing with the remaining 80% being further split into 80% train and 20% validation data. The authors used cross-validation, but the number of folds was not mentioned. Training was carried out over 100 epochs with a batch size of 32, Adam optimizer, and a learning rate of 0.0001. The authors claim 100% accuracy with both models, but it is not specified if this was training, validation or testing accuracy.

Attallah et al. [67] developed a pipeline using three lightweight CNNs (ResNet18, ShuffleNet, and MobileNet) and TL to extract important features. For classification they considered six ML classifiers: NB, KNN, DT, LDA, SVM, and QDA. The dataset included 10 tomato leaf disease classes. The classification was tested in three scenarios: features from each CNN separately, combined features from all three CNNs, and features selected by a hybrid feature selection method. The results showed the best performance, with the KNN and SVM classifiers achieving the highest accuracy of 99.92% and 99.90%, respectively, using only 22 and 24 selected features. Metrics like sensitivity, specificity, precision, F1-score, and MCC all peaked at 99.9% with KNN. The forward search feature selection further improved performance for most classifiers and reduced the feature count while maintaining high accuracy.

Paul et al. [68] introduced a lightweight custom CNN model and compared it to TL-based models (VGG16 and VGG19) for the classification of tomato leaf diseases across 11 classes, including healthy leaves. The study employed data augmentation techniques such as shear, zoom, horizontal/vertical flips, and width/height shifts to improve model robustness. Pre-trained models trained on ImageNet were used with a learning rate of 0.0001, and an ablation study was conducted to optimize model parameters. The custom CNN model achieved the highest accuracy and recall of 95%, outperforming TL-based models.

Ramya et al. [69] utilized a pre-trained AlexNet on a combined dataset made up of PlantVillage tomato images as

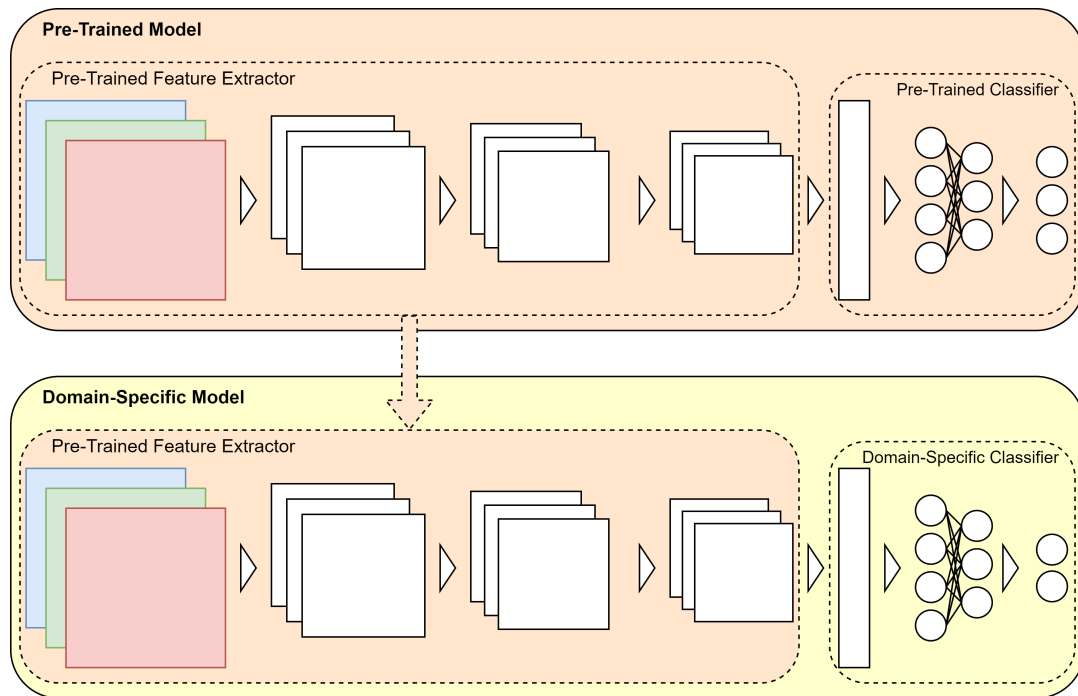


FIGURE 9. Visualization of the way Transfer-Learning works.

well as uncited Kaggle and Mendeley datasets. This resulted in a 10 class dataset with 54,306 total images. The dataset also received augmentation. The applied augmentations were flipping, cropping, rotation, skew, and zoom, before being 80-20 split into train and validation data. Adam, SGDM, cross entropy, and a learning rate of 0.001 were chosen as the hyperparameters. This model achieved 99.7% accuracy on the validation set.

Buchke et al. [70] present an approach for classification of tomato leaf diseases using various versions of EfficientNet applied to the PlantVillage dataset. The dataset includes 10,000 images of 32x32 resolution divided into 10 classes, with 900 images for training and 100 for validation in each class. TL is utilized with experiments conducted on subsets of 3,000, 8,000, and 10,000 images. The model achieves accuracies of 97.3%, 99.2%, and 99.5% for these datasets, respectively.

Radočaj et al. [71] introduce IncMB, which is based on the Inception module, Mish activation function, and Batch Normalization. Using a CNN with TL as the base, three approaches were evaluated for detecting 6 tomato leaf diseases. The public PlantVillage dataset, containing 3,300 images, was split into training and validation sets in a 70:30 ratio, with augmentation techniques such as stretching, mirroring, rotation, contrast enhancement, and brightness adjustments applied to expand the dataset. The best performance was achieved with the pre-trained InceptionV3 network incorporating the IncMB module, attaining an accuracy of 97.78%.

Bouni et al. [72] combine traditional feature engineering with deep learning techniques. The dataset includes 7,311

images of healthy and diseased samples from the PlantVillage dataset, with pre-processing steps like Z-score normalization and augmentation involving color, shape, and transformation adjustments. Regions of interest (ROIs) in the images are segmented and analyzed using handcrafted features based on asymmetry, borders, colors, and diameter (ABCD rule). The extracted features are combined with outputs from a CNN pretrained on ImageNet through mutual information-based feature fusion. This results in a 15% improvement in recognition accuracy and an overall accuracy of 92% during field tests.

Batool et al. [73] suggest a T-Net model trained using Active Learning through Verification of Interesting Knowledge. The ALVIN trained T-Net model is then compared to pre-trained InceptionV3, AlexNet, and VGG16. PlantVillage tomato images are used in this study, which were augmented with shearing, zooming, and flipping and then normalized. Training was carried out with a batch size of 32, a learning rate of 0.01, for 80 epochs, using the Adam optimizer. Results had the VGG16 model reach 96% accuracy, the AlexNet reach 95%, the InceptionV3 model reach 95% and the proposed model reach 97.90%.

b: Papers using Open Lab-Based Grape Leaf Image Datasets

Nagi et al. [74] use PlantVillage to train a grape leaf disease detection model. In their work they trained a pre-trained MobileNet [13] and compared it to pre-trained AlexNet [50], VGG16, and VGG19 [52]. PlantVillage provides 4 classes of grape images, one of which is healthy, totaling 3,423

images, which are resized to 128 x 128 pixels and split into training and validation sets with a ratio of 70-30. The paper examines 3 different optimizers with different learning rates, those being SGD (0.001), RMSProp (0.001) and Adam (0.01), all of which were trained for a total of 30 epochs. SGD achieved 97.18%, RMSProp 97.68% and Adam 98.53% on the validation set. When compared to AlexNet, VGG16, and VGG19, the MobileNet performed the strongest.

Al-Imran *et al.* [75] introduce an advanced approach for identifying grapevine leaf disorders and compare that model with pre-trained EfficientNet-B0, Xception and MobileNetV3 models. The dataset used here was PlantVillage with 4 classes of grape leaf images. Images were split 70-20-10 for train-validation-test and augmented through rotation and zooming. The models were trained using the Adam optimizer over 100 epochs with a batch size of 32. Among the evaluated models, EfficientNet-B0 achieved the highest accuracy at 99.94%, followed by Xception (99.88%), MobileNetV3 (99.45%), and the custom CNN model (99.48%).

Canghai *et al.* [76] presented GC-MobileNet, a lightweight and efficient grape disease recognition model designed for classification and severity grading. Built on MobileNetV3, GC-MobileNet integrates Ghost modules to reduce parameters and enhance feature extraction, and applies the CBAM attention mechanism for improved spatial and channel representation. LeakyReLU is used to better retain features during activation. The model was trained on a dataset of 3,145 grape leaf images (224x224), covering four classes: grape black measles, black rot, leaf blight, and healthy. Data augmentation techniques included random brightness adjustment, normalization, flipping, rotation, and uniform resizing. After testing various configurations, a batch size of 32 and 50 training epochs were selected for optimal performance. GC-MobileNet achieved a classification accuracy of 98.63%, outperforming MobileNetV3 by 6.51% and showing clear advantages over VGG16, ResNet50, and SqueezeNet.

c: Papers using Open Lab-Based Apple Leaf Image Datasets

Pradhan *et al.* [46] compare a total of 10 different ImageNet pre-trained CNN architectures on their apple leaf detection performance. The models chosen for this work were DenseNet201, DenseNet169, InceptionV3, InceptionResNetV2 [58], MobileNet, MobileNetV2, ResNet50, VGG16, VGG19, and Xception. The authors used all 4 apple leaf classes contained in the PlantVillage Dataset, resulting in a dataset of 2,537 images in total. All models were trained for 30 epochs on a training set that makes up 75% of the total data, with validation data making up the other 25%. Results show that in this set of experiments, the DenseNet201 architecture managed the highest accuracy at 98.75%, with all other models besides ResNet50 (74.29%) achieving over 95%. Additionally, the pre-trained feature extracting DenseNet201 was also used in combination with other classifiers such as support vector machines (SVM) and random forest (RF), none of which managed to outperform the 98.75%.

d: Papers using Open Lab-Based Multi-Plant Leaf Image Datasets

Reda *et al.* [77] showcase a mobile application and ML model for plant disease identification. The dataset used for training the backbone was PlantVillage, but a version that includes 39 classes and 61,486. The additional class, when compared to the 38 class PlantVillage set, is a class of only background without a plant in it. The dataset was split 60-20-20 for train, test and validation, without any augmentation made to the already large dataset. The models used during the preliminary stages for testing were pre-trained ImageNet MobileNet, MobileNetV2, NasNetMobile, and EfficientNet-B0, which were trained for 30 epochs with a learning rate of 0.0001, a batch size of 32 and the Adam optimizer. All 4 models were run with 4 different transfer learning approaches, once with all CNN layers frozen, once with 80% of the layers frozen, one with 50%, and one where no layers are frozen. The strongest model-freezing combination was EfficientNet-B0 with no frozen layers, which achieved over 99% accuracy in all test scenarios (preliminary, post-preliminary, k-fold validation).

Bruno *et al.* [78] analyze the performance of the EfficientNet family of CNN models for plant leaf disease recognition (EfficientNet-B0 to B7). Here the 39 class PlantVillage dataset was used, both the augmented and original versions. Images were split 90% for training, 7% for validation, and 3% for testing. All models were loaded with pre-trained weights that were trained on ImageNet and all models were trained in two stages. In the first stage the models were trained with all layers unfrozen, while the second stage froze all CNN layers and only trained the classifier top. Training was carried out until early-stopping (with a patience of 10) found the model to no longer improve. Results show that after stage 1 EfficientNet-B6 (weighted test F1-score 99.7596%) performed best on the original data and EfficientNet-B7 (weighted test F1-score 99.8916 %) performed best on the augmented data. After the second stage, EfficientNet-B7 performed best on both sets with 99.8197% on the original data and 99.8916% on the augmented set. The authors then combined 2 B0 variants (due to their lightweight nature) into an ensemble structure that achieved 100% weighted F1-Score on both test sets.

Eunice *et al.* [79] present the evaluation of the DenseNet121, ResNet50, VGG16, and InceptionV4 models, pre-trained on ImageNet, for their performance on the PlantVillage dataset. Rotation, flipping, zoom, and rescaling were used to augment the dataset after first being pre-processed to the correct size required by each network. Image size was 224x224 for all models except InceptionV4, which required 299x299. Data was then split into train, test, and validation sets by a ratio of 80-10-10. Training was carried out over 30 epochs with Stochastic Gradient Descent and a learning rate of 0.001. Testing has shown that the DenseNet121 architecture has proven to be the best performing with a testing accuracy of 99.81%.

Orchi *et al.* [80] explored methods (SVM, LDA, KNN, CART, RF, and NB) with deep transfer learning models

(VGG16, VGG19, InceptionV3, ResNet50, and CNN). The experiments were conducted on the PlantVillage dataset, which includes healthy and diseased crop leaves for binary classification. The researchers also tested different activation functions and DL optimizers to improve CNN performance. Their results showed that traditional models performed less effectively, with NB achieving the lowest accuracy at 60.09%. In contrast, the DL InceptionV3 model achieved the highest classification accuracy of 98.01%, demonstrating the superior capability of TL approaches for disease detection.

Orlando *et al.* [81] use PlantVillage with all 38 classes for comparative multi-plant disease classification of three models. Three CNN models: DenseNet201, ResNet50, and InceptionV3, were used for image classification. The dataset was split 80% for training and the remaining 20% for testing. DenseNet201 and InceptionV3 achieved the highest accuracy of 98%, slightly outperforming ResNet50, which achieved 97%.

Hari *et al.* [82] propose a lightweight and efficient deep learning model for plant leaf disease classification. The model incorporates a feature reuse strategy. Trained on a dataset combined from Kaggle and Mendeley sources of banana, guava and mango fruit crops (70-15-15 split and augmented with rotation, and shifting) it achieves a high accuracy of 99.14% for identifying diseased leaves, surpassing 15 pre-trained models, including VGG16, ResNet50, InceptionV3, DenseNet121, and MobileNetV2, in both accuracy and computational efficiency.

Sood *et al.* [83] present a grape leaf diseases classification model using the PlantVillage dataset (grape classes only) containing 4,063 images. The dataset was split into 80% for training and 20% for validation. Images were resized to 224×224 for most architectures and 299×299 for InceptionV3. The proposed CNN model was compared to VGG16, ResNet50, InceptionV3, and DenseNet121. Using 150 epochs, a batch size of 16, and the Adam optimizer, the proposed model achieved an impressive accuracy of 99.88%, outperforming pre-trained models such as DenseNet121, which achieved 98.65%.

Lu *et al.* [84] compare EfficientNet-B5, MobileNet, ResNet50, InceptionV3, and VGG16 for the field of plant leaf disease classification. The dataset was a what seems to be an augmented version of the PlantVillage dataset, resulting in 38 classes with a total of 87,867 images (split 80-18-2). TL was employed, where pre-trained models served as the foundation and were then fine-tuned using said dataset. Among the tested models, EfficientNet-B5 demonstrated the best performance, achieving a classification accuracy of 99.2%. Training was conducted with hyperparameters of batch size 32 for 15 epochs with a learning rate of 0.00001, and the Adam optimizer. The input size was 224×224 .

Quan *et al.* [85] propose the MS-Net model, which is an modified version of MobileNetV3. Models were pre-trained on the Pl@ntNet-300k dataset before being train on the PlantVillage and the FGVC7 datasets. The pre-tarin mechanisms of PlantNet-300k was compared with ImageNet pre-

training for TL with FGVC7 but not PlantVillage. Results with Pl@ntNet-300k were shown to be better performing than ImageNet trained models. On the FGVC7 dataset, the propose model performed best (97.70%), with Xception being the best of the remaining models (95.58%). In PlantVillage the proposed model managed 99.80% with Xception coming in second with 99.74%.

Bhagatet *et al.* [86] focus on tomato, potato, and bell pepper leaves, using PlantVillage and other image data, resulting in 18,165 images split 80-20. Xception and VGG16 were trained on this data and the results were compared. The best performing model was optimized using the Adam optimizer and evaluated using metrics such as validation accuracy, loss, F1-Score, precision, recall, AUC, and ROC. The Adam optimized and learning rate 0.0001 VGG16 outperformed other models, achieving training and validation accuracies of 98.42% and 91.85% for dataset 1, 100% and 97.21% for dataset 2, and 99.75% and 98.18% for dataset 3.

Uskaner *et al.* [23] proposed a few-shot learning framework. Their framework leverages a pre-trained model on the ImageNet dataset, fine-tuned on the PlantCLEF2022 dataset for extracting plant-based features. Plant leaf image embeddings are generated using this fine-tuned model, and the CNN model's classification head is replaced with a SVM classifier, enabling effective training with fewer example images per class. Experiments were conducted on the PlantVillage dataset (38 classes) and the PDD271 dataset (271 classes). On the PlantVillage dataset, the framework achieved average accuracies of 88.4% at 10-shots and 75.5% at 5-shots. For the PDD271 dataset, which presents a larger number of classes, the framework provided accuracies of 56.3% for single-shot, 67.5% for 3-shots, and 74.2% for 5-shots.

Shafik *et al.* [87] introduce two advanced Plant Disease Detection Network (PDDNet) models: PDDNet-AE (Early Fusion) and PDDNet-LVE (Lead Voting Ensemble). These models integrate nine pre-trained Convolutional Neural Networks (CNNs), including DenseNet201, ResNet101, ResNet50, GoogLeNet, AlexNet, ResNet18, EfficientNet-B7, NAS-NetMobile, and ConvNeXtSmall, fine-tuned through deep feature extraction for efficient plant disease identification and classification. The experiments utilized the PlantVillage dataset, with this study focusing on 15 categories of disease-specific images processed in three formats: color, gray-scaled, and segmented. Data augmentation techniques, including noise injection, scaling, flipping, rotation, and color adjustments, were employed to enhance the dataset and improve model generalization. Hyperparameter tuning involved varying batch sizes from 10 to 100, saturating at 10 epochs, with a learning rate range between 0.1 and 0.001. A gradient threshold of 1 was applied to stabilize training. PDDNet-AE achieved an accuracy of 96.74%, while PDDNet-LVE outperformed with 97.79%, significantly exceeding the performance of standalone CNN.

Chilakalapudi *et al.* [88] proposes an automated framework using the Chronological Flamingo Search Algorithm (CFSA) combined with TL. The process begins with pre-

processing leaf images using Adaptive Anisotropic Diffusion to remove noise, followed by data augmentation techniques such as padding, rotation, translation, transformation, contrast adjustment, hue modification, and saturation balancing. For segmentation, U-Net++ is employed to isolate diseased leaf regions. After feature extraction was performed to gather essential information, it was followed by a two-level classification: plant type at the first level and disease category at the second. Classification uses a CNN integrated with TL, leveraging the LeNet architecture and trained via CFSA using PlantVillage. The model was trained with a batch size of 128, kernel size 5×5 , categorical cross-entropy as the loss function, and a learning rate of 0.5. The proposed CFSA-TL-based CNN with LeNet achieved an accuracy of 95.7%, sensitivity of 96.5%, and specificity of 94.7%.

Dheeraj et al. [89] introduce a novel Lightweight DenseNet (LWDN) for efficient plant disease detection, addressing the computational challenges of traditional CNNs, built on DenseNet121. The model was trained on the 39 class PlantVillage dataset. It achieved 99.37% accuracy (while being 93% smaller than InceptionV3 and Xception). Training in this paper was carried out with a learning rate of 0.0001, Adam optimizer, batch size 16, dropout 0.5, and 50 epochs.

Cosma et al. [90] proposed to fix the imbalance in the PlantVillage dataset, using 10 classes (only pepper, potato, and tomato), with augmentation techniques that were applied to generate a balanced training set. A custom CNN model was developed using transfer learning with ImageNet-pretrained weights and compared against eight advanced models. Models were trained with a batch size of 32, Adam optimizer, 100 epochs, and a learning rate of 0.001 (AlexNet used 0.0005). Despite limited data, the proposed CNN achieved 98.2% accuracy, outperforming existing models by 3.98% to 16.86%.

Oubadriss et al. [91] utilized the full PlantVillage dataset with 54,201 images of plant leaves with all 14 species and 38 disease classes. Two pre-trained CNN models, VGG16 and ResNet50, were evaluated for leaf disease classification. Both models were trained for 8 epochs. Performance was measured using classification accuracy and F1-Score, with VGG-16 achieving 97.30% accuracy, outperforming ResNet50's 93.04%.

Alramli et al. [92] introduce a hybrid model combining Efficient Neural Network (ENN) and Squeeze-and-Excitation Network (SEN) for plant disease classification. Using U-Net segmentation, patch creation, and denoising, the model was trained with transfer learning, cross-channel attention, and hyperparameter optimization to prevent overfitting. Evaluated on PlantVillage (54,306 images, 40 classes) and Rose Leaf Disease (14,910 images) datasets, it achieved 96.40% and 97.15% accuracy respectively, outperforming VGG16, InceptionV3, and ResNet50. Training ran for 35 epochs with optimized settings: input 224×224 images, batch size 64, learning-rate 0.001, and dropout 0.3, offering superior performance with lower complexity.

Kafare et al. [93] suggested TL-PDD framework for plant disease detection, integrating pre-processing, segmentation,

feature extraction, and prediction. Median filtering is used for image refinement, and APIJS segmentation isolates disease regions. Key features are extracted using Multi-texton, PHOG, and NMA-LGIP techniques. Disease classification is performed using the proposed DCA-CNN-TL model, which leverages double convolutional activation with transfer learning. Their model was trained on a dataset of 7,231 images (grape: 4,062; apple: 3,169) across 8 disease classes, with 224×224 input size, batch size 32, and 25 epochs. The proposed, DCA-CNN-TL model achieved a specificity of 0.916, outperforming baseline models like SVM (0.882), LSTM (0.885), and CNN+TL (0.888), showing high precision in identifying plant diseases.

e: Papers using Open Lab-Based Corn/Maize Leaf Image Datasets

Mafukidze et al. [47] use InceptionV3, DenseNet121, DenseNet201, MobileNetV2, VGG16, EfficientNetB0, ResNet50, and LeNet [49] for maize leaf classification. Maize images were taken from the PlantVillage dataset and augmented via shearing, rotation, flipping, saturation, mean filtering, mean filtering + flipping, and cropping. This process increased the total image count from 4,188 to 46,898 images after augmentation. Images were split 80-20 into train and test subsets and then trained with each of the models for a total of 500 epochs. Results indicate that InceptionV3, DenseNet121, DenseNet201, MobileNetV2, VGG16, EfficientNetB0, and ResNet50 all managed a 99% overall accuracy, with LeNet reaching 94%. The authors also implemented their own CNN models, but those were not trained with transfer-learning.

An et al. [94] took the PlantVillage dataset and extracted only the corn images. This results in a dataset of 4 classes, one of which is healthy, that is then augmented via flipping and translations. Images are resized to 227×227 and then split 80-20 for training and validation. The model that this data is then used for is AlexNet, which is trained for 20 epochs with cross-entropy loss, stochastic gradient descent (SGD) optimizer and also Adam optimizer, learning rate 0.0001, and a batch size of 124. The model reached 95.7% with SGD, which managed to outperform the same setup with Adam with 95%.

f: Papers using Open Lab-Based Rice Leaf Image Datasets

Latif et al. [95] use GoogleNet, VGG16, VGG19, DenseNet201, and AlexNet on a rice leaf dataset. The dataset contains 6 classes for a total of 2,164 images before augmentation (rotation, translation, and scale-in/scale-out). Before training, the images were also pre-processed with edge-aware local contrast, smoothing, and Fourier transform, before being split into 80-20 train-test sets. Training was carried out 3 times, once with non normalized original images as input, once with non-normalized augmented images and once with the augmented dataset after normalization. The highest accuracy was achieved by VGG19 on the non-normalized augmented dataset with 96.08% accuracy.

Simhadri et al. [96] studied rice leaf disease classification with a focus on using transfer learning of 15 pre-trained CNN

models (Efficientnet-B0, InceptionV3, InceptionresnetV2, ResNet50, DenseNet201, Xception, ResNet101, Nasnet-mobile, MobilenetV2, Darknet53, ShuffleNet, GoogLeNet, SqueezeNet, VGG16, and AlexNet) to automatically identify rice leaf diseases. Among these models, InceptionV3 achieved the best performance, with an average accuracy of 99.64% and strong metrics across the board: precision (98.23%), recall (98.21%), F1-Score (98.20%), and specificity (99.80%). In contrast, the AlexNet model showed the lowest accuracy of 97.35%.

Abasi et al. [97] present a comparative study of InceptionV3 and EfficientNet-B2 and a proposed model on a small scale 6 class rice leaf disease image dataset. The dataset consisted of only 256 training images and 64 validation images. The training process used the Adam optimizer with a categorical cross-entropy loss function, a batch size of 32, and a learning rate of 0.001. Early stopping, based on validation loss, was implemented to ensure optimal performance. The model was trained for 30 epochs and 100 for fine-tuning and managed an accuracy score of 91.4%, with the other models not managing such high performance.

Kongara et al. [98] classified paddy crop diseases using transfer learning and image processing techniques. Their dataset, sourced from Kaggle, contains 2,000 images across 4 disease classes, each resized to 256×256 pixels. Augmentation was carried out through rotation, flipping, and normalization. Several pretrained models, including InceptionV3, VGG16, and ResNet50, were evaluated, with InceptionV3 achieving the highest accuracy of 91.23%. The model was trained for 30 epochs using standard deep learning procedures to assess classification performance.

g: Papers using Open Lab-Based Coffee Leaf Image Datasets

Novtahaning et al. [99] analyze the performance of a number of different CNN models on a coffee plant leaf disease dataset [100]. The dataset consists of images taken in lab in Brazil and contains 1,300 across 5 classes. Models considered in this work are EfficientNet-B0, ResNet152, VGG16, InceptionV3, Xception, MobileNetV2, DenseNet201, InceptionResNetV2, and NasNetMobile. The best 3 models were then combined into an ensemble arrangement to create a new model also tested on the same data. Training was conducted over 50 epochs with 3 optimizers, in order to find the most suitable one. SGD, Adam, and RMSProp were considered. Adam proved to be the strongest one. As for models, the 3 best performing models were EfficientNet-B0, VGG16, and ResNet152. After fine-tuning they managed 95%, 94.2%, and 93.8% respectively. The ensemble model managed to outperform the single model approaches with 97.3% on the validation data.

h: Papers using Open Lab-Based Olive Leaf Image Datasets

Diker et al. [43] mainly focus on the use of more traditional machine learning methods as classifiers in combination with pre-trained CNN. The CNN (ResNet101 and MobilNet) are used as features extractors, which then forward these features

to the classifiers, namely random forest (RF), bagged RF, support vector machine (SVM), and also a rather deep artificial neural network (ANN) with 20 layers. The data was collected by the authors in Turkey under lab conditions and made public as part of the paper's contributions. In total 954 images were collected and labeled into 2 classes (healthy and infected). The background was removed with masking and then the images were resized to 224×224 . Training was conducted with the k-fold approach, with 5 and 10 folds for all feature extractor and classifier combinations. After training, the best results were obtained via the 10 fold SVM in combination with ResNet101. The best results with the ANN classifier were also obtained with ResNet101, but with 10 fold cross-validation.

i: Papers using Open Lab-Based Potato Leaf Image Datasets

Lanjewar et al. [101] made use of pre-trained NASNetMobile, VGG19, and DenseNet169, ResNet50V2, InceptionV3, and Xception models in their work. An open Kaggle dataset for potato disease images (early blight, late blight, healthy) with 1,500 images was used to train the models, after applying augmentation methods (rotation and flip). The data was split 60-20-20, 60% being the training data and resized to 224×224 . Models were trained for 100 epochs with the Adam optimizer and the categorical cross-entropy loss function, as well as a batch size of 32, after which the DenseNet model reached the highest accuracy score with 99.67% on the test set. The other models achieved 97.00% (InceptionV3 and ResNet50V2), 96.00% (Xception), 95.67% (VGG19), and 95.33% (NASNetMobile).

Shaheed et al. [102] mainly focus on a newly developed model. Said model is, however, compared to other approaches such as pre-trained ResNet50, VGG16, Xception, and InceptionV3 and Swin-Transformers. These models were trained on the PlantVillage potato subset, after it was extended via augmentation methods (rotating, shearing, shifting), resulting in 3,600 images that were split with a 80-20 ratio. Results have shown that, while the proposed model managed the highest accuracy score on the validation data, the Inception net was second best, and not far behind, with 98.09%, outperforming the Swin-Transformer ViT.

Bania et al. [103] combine Principal Component Analysis (PCA) with ResNet101 to train on a Kaggle potato leaf dataset with 3 classes and 1,800 images that were increased to 2,152 with flipping, rotation, and translation augmentation. Images were also contrast enhanced and split 80-20 with 80% of the data being used for the training of the model and the remaining 20% making up the testing subset. The model managed 95.13% training on the dataset.

j: Papers using Open Lab-Based Wheat Leaf Image Datasets

Reiset al. [104] propose an Integrated Deep Learning Framework (IDLF) combined with ensemble learning (EL) to classify wheat plant diseases. They employ pre-trained deep neu-

ral networks enhanced with techniques like hyper-column, contrast stretching, and Contrast Limited Adaptive Histogram Equalization (CLAHE). Training was carried out with Adam optimization, categorical cross-entropy loss, a learning rate of 0.000001, and early stopping with validation monitoring, including 30 epochs for all experiments. Their comparative results showcase RegNetY080 achieving 97.64% accuracy when trained from scratch, 98.33% with CLAHE optimization, and 99.58% when combined with transfer learning and Random Forest. The hybrid IDLF and EL model achieved 99.72% accuracy. Applied to wheat datasets, it reached 97.56% for Wheat Leaf and 99.43% for Wheat Leaf Disease datasets. The Xception model performed best with 98.06% accuracy, while NASNetMobile underperformed at 48.61%.

2) Papers using Open Field-Collected Image Datasets

a: Papers using Open Field-Collected Pear Leaf Image Datasets

Alirezazadeh et al. [105] create a Convolutional Block Attention Module (CBAM) enhanced model for pear leaf disease identification using the open DiaMOS dataset [29]. The DiaMOS dataset was collected in Italy and consists of images taken in field. A total of 4 classes (healthy, spot, curl, slug) are present, which amount to 3,006 images overall. Augmentation techniques were also applied (mirroring, rotation, and color variation). The models compared in this paper are EfficientNet-B0, MobileNetV2, ResNet50, InceptionV3, and VGG19, all of which were trained once without CBAM and once with. The CBAM block directly follows the last layer of the feature extracting CNN and before the fully connected classifier. The models were trained for 100 epochs and results shows that CBAM improves the performance of all models, with the EfficientNet-B0 being the strongest in both categories. EfficientNet-B0 without CBAM achieved 85.82% and EfficientNet-B0 without CBAM with CBAM reached 86.89% with 10-fold cross-validation.

b: Papers using Open Field-Collected Enset Leaf Image Datasets

Fulle et al. [44] present work that focuses on the classification of enset images. Enset is a crop often used in Ethiopia, so the authors collected their own binary field Enset leaf image dataset which they made public through Kaggle and it contains a total of 1,228 images. In this work the authors trained a total of 4 CNN models with pre-trained weights from ImageNet. These 4 models were MobileNetV3-Small, NASNetMobile, MobileNetV2, as well as DenseNet121. To provide the models with more data, augmentation techniques were applied to the dataset to increase its size to 7,580. To achieve that shearing, flipping, and brightness changes were applied. The networks were each trained for 30 epochs with the Adam optimizer, a learning rate of 0.0001, a batch size of 64, and the binary cross entropy loss function. Data was split into a training set with 70% a validation set with 10% and a test set with 20%. When looking at the trained models performance on the testing set we can see that the pretrained

MobileNetV3 was the strongest model with 99.93%. Both MobileNetV2 and DenseNet121 also managed above 94% with NASNet Mobile only managing 76.68%.

c: Papers using Open Field-Collected Rice Leaf Image Datasets

Mavaddat et al. [106] test VGG16, InceptionV3, and Resnet152v2 on a open rice leaf disease dataset. The dataset [24] features 4 classes (bacterial blight, blast, brown spot, and tungro) totaling 5,932 images. All models were pre-trained on the ImageNet dataset before being adjusted to the 4 classes use case in this transfer-learning application. The models were configured in 2 different ways. Once models had all CNN layers frozen with only the classifier being trained and the other approach left the last set of convolutional layers unfrozen to be trained alongside the classifier top. Freezing the CNN completely was only used in combination with VGG16, while the second approach was done with all models. Both approaches also used different hyperparameters, with the fully frozen approach using a batch size of 64, the RM-Sprop optimizer, a learning rate 0.0001, and cross entropy loss, while the other method uses a 32 batch size, the Adam optimizer, a learning rate of 0.001, and the same cross entropy loss. Results show that all 3 models managed, when looking at their best results with the best configuration for each model, 100% accuracy.

Mohapatra et al. [107] have implemented and trained a pre-trained AlexNet model for rice leaf disease detection. The dataset contains 3 classes (bacterial leaf blight, brown spot, and leaf smut), which add up to 1,732 images, but the dataset itself was not referenced. Based on the fact that the pre-processing includes background removal, one can assume that the dataset is a field dataset. Then, before training, the data was split into 90% training data, and 5% each for validation and testing. The AlexNet's pre-trained weights have been obtained from training on the ImageNet dataset. After transfer learning on the new dataset, the AlexNet model managed to score 98.80% accuracy.

Sudhesh et al. [108] focused on identifying rice leaf diseases using VGG16, VGG19, ResNet50, ResNet101, InceptionV3, XceptionNet, DenseNet121, InceptionResNet, MobileNet, and NASNetMobile. The researchers worked with a dataset of 3,416 images with 4 classes. They tested 10 pre-trained deep learning models and found DenseNet121 achieved the highest accuracy (93.87%) among them. Additionally, they used features from these models' final layers to train three ML classifiers and discovered that the Random Forest classifier paired with DenseNet121 features performed better than other combinations. To further improve results, they proposed a Dynamic Mode Decomposition (DMD) based attention-driven pre-processing method, which localized infected areas by generating segmentation maps. Using this pre-processing, the XceptionNet model with an SVM classifier achieved a perfect test accuracy of 100%. On real-world rice leaf images, Xception achieved 94.33% accuracy, outperforming other models. Metrics like precision,

recall, and F1-score showed that models trained with DMD-preprocessed images performed significantly better overall.

Bruno et al. [109] concentrate on developing an efficient method to detect rice plant diseases, using a dataset with two common rice diseases, alongside healthy samples. Their dataset included 7,508 images, categorized into blast (2,988 images), brown (2,988 images), and healthy (1,532 images). Data augmentation techniques, such as rotation, shifting, flipping, brightness adjustment, and zoom, were applied to enhance model performance. The model was trained for 100 epochs and achieved a 99% accuracy rate after training was completed.

d: Papers using Open Field-Collected Cassava Leaf Image Datasets

Paiva et al. [110] use the iCassava dataset [30], to train, test and evaluate a total of 12 different pre-trained CNN architectures. The CNN models that were chosen for this work are VGG16, VGG19, ResNet50, ResNet50V2, ResNet101V2, ResNet152V2, InceptionV3, InceptionResNetV2, MobileNetV2, DenseNet121, DenseNet169, and DenseNet201. The dataset, as it was used in this work, comes with 21,397 that span across 5 total classes, one of which is healthy leaves. An 80-10-10 train-test-validation split was applied to the data prior to training. Additionally, augmentation techniques such as rotation, brightness changes, zooming, and flipping, were applied. Training was then conducted with 32 images per batch, Adam optimizer, differing learning rate for each model, all of that for 20 epochs, but with early stopping (patience 7) enabled. Results proved that DenseNet169 performed the best with 74.77% on the test set.

Ravi et al. [111] propose an attention-enhanced approach using pretrained EfficientNet models (A_EfficientNet-B4, A_EfficientNet-B5, and A_EfficientNet-B6) for Cassava leaf disease classification. Features from the penultimate layers were extracted, reduced using kernel principal component analysis (KPCA), and fused for classification using a stacked ensemble meta-classifier combining random forest, SVM, and logistic regression. Experiments were conducted on a publicly available Cassava leaf disease dataset (iCassava) containing 10,839 images across five classes (blight, streak, green mottle, mosaic, healthy). Training was performed with the Adam optimizer, a learning rate of 0.0001, a batch size of 64, and 30 epochs. The proposed method achieved an overall accuracy of 88%, with over 95% accuracy for each individual class.

e: Papers using Open Field-Collected Coffee Leaf Image Datasets

Somanna et al. [112] evaluate the effectiveness of machine learning techniques, particularly DL and transfer learning models, for detecting diseases in Arabica coffee leaves. Using the "JMuBEN" dataset, comprising 58,405 images categorized into five classes (four disease types: Phoma, Cercospora, Leaf Rust, Miner, and healthy leaves), various models such as EfficientNet-B0, MobileNetV2, CNN, and

VGG16 were assessed for their performance in addressing the multi-class label problem. EfficientNet-B0 emerged as the best-performing model, achieving a test accuracy of 99.72% with an F1-Score, recall, and precision of 99.7%.

f: Papers using Open Field-Collected Chili Leaf Image Datasets

Li et al. [113] address the challenges of chili leaf diseases classification by proposing a model named MCCM (MCSAM-ConvNeXt-MSFFM). They integrate ConvNeXt with a Multi-Scale Feature Fusion Module (MSFFM) to better capture disease features of varying sizes and positions and incorporates a Mixed Channel Spatial Attention Mechanism (MCSAM) to enhance spatial and channel correlations. Pre-trained weights from the PlantVillage dataset are used for TL to accelerate training. The dataset includes 2,500 augmented images of five categories: healthy chili leaves, leaf curl disease, leaf spot disease, whitefly disease, and yellowing disease, resized to 224×224 pixels. Augmentations such as Gaussian noise, random brightness, rotation, and occlusion further enrich the dataset. The MCCM model surpasses existing CNN models (VGG16, ResNet34, GoogLeNet, MobileNetV2, ShuffleNet, EfficientNetV2, ConvNeXt) and Swin-Transformer, achieving average improvements of 3.38% in accuracy, 2.62% in precision, 2.48% in recall, and 2.53% in the F1 score.

g: Papers using Open Field-Collected Multi-Plant Leaf Image Datasets

Mazumder et al. [114] evaluated, as part of their study, an attention enhanced extended DenseNet201, and compared it with MobileNetV2, DenseNet121, ResNet152V2, DenseNet169, DenseNet201, InceptionV3, NASNetLarge, InceptionResNetV2, EfficientNetV2S, and EfficientNetV2L models. The datasets used are a banana dataset [38] and a black gram dataset [39], that were treated separately. The images were normalized to be within the range of 0 and 1 for each pixel and resized to 224x224. The data was also augmented via rotation, flipping, scaling, cropping, and noise. The Adam optimizer was used with a small batch size of only 8. The learning rate was set to 0.00001 and the models were all trained for 50 epochs with early stopping enabled. The best basic CNN (MobileNetV2) reached 93% with the proposed model managing 99,50%.

h: Papers using Open Field-Collected Mango Leaf Image Datasets

Singh et al. [115] present a Deep Transfer Learning Driven (DTLD) model for mango leaf disease detection, leveraging VGG16 and DenseNet121 to improve classification accuracy. The dataset consists of 4,000 images across seven disease classes and a healthy category, with pre-processing steps including resizing (240×240), rescaling, flipping, and rotation for augmentation. The data is split into training, validation, and testing sets. The model uses the softmax activation function for classification and is trained with the Adam optimizer

(learning rate 0.001, β_1 0.9, β_2 0.999). The proposed model achieves 99.76% accuracy in the end.

i: Papers using Open Field-Collected Tomato Leaf Image Datasets

Catal et al. [116] introduced a multi-deep and meta-learning fusion framework for automatic tomato leaf disease detection, classifying images into healthy, early blight, and late blight categories on a open tomato leaf dataset. The proposed approach integrates CNN, transformer, and MLP-based models pretrained on ImageNet with a voting classifier (VtC) for enhanced accuracy. Hybrid models incorporate CoAtNet, FasterNet, FastViT, WaveMLP, and DenseNet201, while individual architectures include DenseNet201, MobileNet, LCNet, GhostNet, BeiT, FlexiViT, MaxViT, FastViT, MLP Mixer, and WaveMLP. The dataset, improved using CLAHE, consists of 224×224 images, split 70% for training and 30% for testing, with a batch size of 16 and was trained for 40 epochs. The model is trained with the Adam optimizer, categorical crossentropy loss, and softmax activation. The proposed framework achieved 99.56% accuracy.

j: Papers using Open Field-Collected Palm Leaf Image Datasets

Hau et al. [117] focuses on detecting palm leaf diseases using vision techniques. The dataset, sourced from the Kaggle "Date Palm data" [42] includes 3,609 images of palm leaves with three classes. Data pre-processing involved augmentation techniques like rotation, shifting, and horizontal flipping to balance the dataset. They employed CNNs and TL using ResNet152V2, VGG19, DenseNet201, MobileNetV2, and InceptionResNetV2. These models demonstrated high efficiency, achieving accuracies above 95%. Additionally, precision, recall, and F1-scores exceeded 90% across all classes.

3) Papers using Open Hybrid Condition Image Datasets

a: Papers using Open Hybrid Condition Corn/Maize Leaf Image Datasets

Albahli et al. [118] combine various corn leaf different datasets, including PlantVillage, plantDoc [17], CD&S [27], and the CornDataset [119], into one to use for training and testing. The resulting dataset contains 8,231 images across 5 classes (healthy, common rust, gray leaf spot, northern leaf blight, southern corn leaf blight) before augmentation. To further enhance the dataset, rotation, flipping, translation, scaling, and color adjustments were used. Images were sized to be 240×240 and then split 7-3 for train and testing sets. The models used in this work are AlexNet, VGG19, InceptionV3, ResNet50, DenseNet201, MobileNetV2, EfficientNet, and EfficientNetV2, as well as a model proposed by the authors by the name of EANet. EANet uses the EfficientNetV2 architecture as the base model upon which EANet was built by addition of spacial attention and channel attention mechanisms. All models were provided with pre-trained weights from ImageNet before training. Training results showed that the new EANet managed to outperform all other models on

the training set with a testing accuracy of 98.94%. All other models achieved scores between 84% and 95%.

b: Papers using Open Hybrid Condition Rice Leaf Image Datasets

Stephen et al. [120] introduce an approach that integrates a 3D2D CNN for feature extraction and an Improved Backtracking Search (IBS)-optimized Deep Generative Adversarial Network (DGAN) for classification. The dataset, comprising 5,200 rice leaf images sourced from Kaggle and Google Images, includes four classes. The images were pre-processed by rescaling pixel values (0–1) and resizing them to 224×224 before training. Models were trained using a learning rate of 0.001, a batch size of 100, and for 100 epochs. The proposed approach achieved classification accuracy, with bacterial blight (98.5%), leaf blast (98.9%), brown spot (99.3%), and healthy leaves (99.7%). Overall, the model outperformed traditional methods such as XGBoost [121], transfer learning, and SVM [122], achieving a 99.25% accuracy, along with 98.25% precision, 98.50% recall, and a 97.75% F-measure.

Aggarwal et al. [123] studied a federated learning (FL) approach considering data privacy and heterogeneous training environments. The Rice-leaf dataset [24] was used for transfer learning with federated averaging strategy with 4 clients. The dataset was divided in both IID (Independent and identically distributed) and non-IID for performance evaluation with heterogeneous data distribution. The authors trained and evaluated DenseNet201, EfficientNet-B3, InceptionNetResNetV2, MobileNetV2, VGG16, VGG19, Xception, and ResNet152V2 along with a Dense Neural Network on FL client's site. For the IID dataset, EfficientNet-B3 achieves 99% training and validation accuracy, while MobileNetV2 records 98% training and 90% validation accuracy. On the non-IID dataset, EfficientNet-B3 attains 100% training and 99% validation accuracy, with MobileNetV2 achieving 99% training and 97% validation accuracy. Results show that the EfficientNet-B3 and MobileNetV2 models reach 99% accuracy on IID data and 100% training with 99% validation accuracy on non-IID data.

Haikal et al. [124] use the paddyDoc [37] dataset, both the in field images as well as the lab images, but both sets are treated separately. The in field dataset of 7,799 images was split 75-20-5 (train-val-test) and the lab dataset contains 3,000 images with a 66-27-7 split. Datasets are augmented in two different ways. Mix based augmentation, including cutout, mixup, cutmix, and fmix were applied and compared with traditional augmentation methods, those being rotation, flipping, color adjustment (brightness, contrast, saturation, and hue), and cropping. To test which augmentation method can generate the best results, ImageNet pre-trained GoogleNet, ResNet34, and MobileNetV3 were employed. The models were trained for 100 epochs with a learning rate of 0.001, the cross entropy loss, SGD optimizer, and a batch size of 32. On the field data, the mix-based augmentation methods generated the strongest models, but it was not as successful on the lab dataset. The overall best model on lab data was GoogleNet

with traditional augmentation with 93.30% accuracy on the test data, while GoogleNet with FMix and MobileNetV3 with all augmentation techniques combined managed 99.75% on the field dataset.

c: Papers using Open Hybrid Condition Multi-Plant Leaf Image Datasets

Sahu et al. [125] proposed a Hybrid Xception TL with Crossover Optimized Kernel Extreme Learning Machine (HXTL-COKELM) for plant leaf disease classification. The approach utilizes the Xception model for feature extraction through TL. Additionally, a crossover-based Tasmanian Devil Optimization (TDO) algorithm optimizes the Kernel Extreme Learning Machine (KELM) parameters for improved classification. The dataset consists of 43,466 images of healthy and diseased plant leaves, pre-processed to a size of 224×224 pixels. The data was divided into 70% for training, 20% for validation, and 10% for testing, with augmentations like re-sizing, normalization, and conversion applied. The proposed HXTL-COKELM method achieved an impressive accuracy of 98.9%.

Joseph et al. [126] address challenges of real time datasets collection for rice, wheat, and maize, encompassing common bacterial and fungal diseases that significantly impact crop yields. The developed datasets were applied to 8 fine-tuned DL models, using consistent hyperparameters, over 100 training epochs. Data augmentations such as horizontal and vertical flipping, 90-degree rotations, height and width shifts, zooming, shearing, and brightness adjustments were employed to enhance model robustness. Among the models, Xception and MobileNet achieved testing accuracies of 95.80% and 94.64%, respectively, for maize leaf disease recognition. For wheat, MobileNetV2 and MobileNet excelled with testing accuracies of 96.32% and 96.28%. In rice leaf disease recognition, Xception and InceptionV3 stood out with accuracies of 97.28% and 96.20%, respectively. The new CNN model trained from scratch managed testing accuracies of 97.04%, 97.06%, and 98.08% for maize, rice, and wheat, respectively. On the wheat leaf dataset, models such as MobileNetV2 and InceptionResNetV2 showed validation accuracies above 90%, with MobileNetV2 achieving the highest testing accuracy of 96.32%.

Shafik et al. [127] propose a hybrid Inception-X (IX-CNN) model for early plant disease detection and classification, combining inception and depthwise separable convolutions to enhance multi-scale feature extraction while reducing complexity and overfitting. The IX-CNN was trained on 6 plant and leaf disease datasets, totaling 80,782 images with an input size of 224×224 . It achieved 100% accuracy on PlantVillage, PlantDoc, and Turkey Disease datasets, and 99.79%, 99.95%, and 98.64% on Rice Disease, RoCole, and NLB, respectively. The model was also deployed as a real-time AI application across MATLAB, Android, and Servlet platforms for an automated leaf disease detection.

Sherly et al. [128] propose a combined DL system for plant species identification and disease detection using healthy and

diseased leaf images. An integrated dataset of 8,000 images was formed by merging 4 datasets, with all images resized to 100×100 pixels and training run for 30 epochs. Two hybrid architectures were evaluated: a Convolutional Siamese Network (CSN) for low-sample training scenarios and a CNN with ensemble classifiers, using both a simple CNN model (SM) and VGG16. Experiments varied training settings, including transfer learning, custom architectures, image types (color, grayscale, segmented), and train-test splits (80:20). Results showed that the multi-learner CNN model improved accuracy by 13.4% (VGG16) and 37.39% (SM) without requiring data augmentation. The highest accuracy of 99.34% was achieved on the PlantVillage dataset using GoogLeNet with TL.

B. PAPERS USING PRIVATE IMAGE DATASETS

In this section the papers that use datasets that are not openly available will be discussed. This means that most of the papers listed here are papers that have collected their own data and where the authors have chosen to keep the data private. In the following sections, we discuss each category consecutively.

1) Papers using Private Lab-Based Image Datasets

a: Papers using Private Lab-Based Multi-Plant Leaf Image Datasets

Tej et al. [129] used a self compiled dataset in their work. The authors collected their own data, which contains diseases that are commonly found in Tunisia. The images were taken under lab conditions. The original dataset contains 488 images, which were extended to 8,200 via data augmentation. This paper uses the ResNet152 and ResNet50 [53] models, both pre-trained on ImageNet [21]. Both models are trained once with the original dataset (488 images) and also with the augmented dataset. When looking at the results (test accuracy), ResNet50 achieved 92.7% on the augmented dataset and 83.3% on the original, while ResNet152 achieved 98.85% on the augmented data and 92% on the original.

b: Papers using Private Lab-Based Papaya Leaf Image Datasets

Bacus et al. [130] utilized the MobileNet architecture [13] combined with transfer learning to classify papaya leaf diseases. The authors fine-tuned the classification layers of the pre-trained model while freezing the earlier layers and subsequently tested the model with unknown samples. Their model was trained on images of six categories. The system was implemented on a Raspberry Pi running LineageOS as the primary hardware platform, capable of processing a dataset of 72 samples and achieving an overall accuracy of 91.66%. Individual class accuracies were reported as 92.3% for blackspot, 75% for brownspot and mealybug infection, and 100% for powdery mildew and healthy papaya leaves.

c: Papers using Private Lab-Based Chili Leaf Image Datasets

Naik et al. [131] collected a dataset of 5,100 chili plant leaf images that were taken in India. All images are taken under lab conditions, spanning across a total of 6 classes, 1 healthy and 5 sick. Data was doubled in size through augmentation and then split into train and validation data 90-10. All models (AlexNet, DarkNet53 [56], DenseNet201, EfficientNet-B0, InceptionV3, MobileNetV2, NasNetLarge [54], ResNet101, ShuffleNet, SqueezeNet [62], VGG19, and Xception [64]) were trained with the augmented data as well as with the original unaltered data for 30 epochs each. The authors also introduce a new model SECNN, which does not use transfer-learning. The paper's validation results indicate that VGG performed the best of the TL models on the dataset without augmentation with 83.54%, while the DarkNet53 model achieved 98.82% (the best results) on the augmented data.

2) Papers using Private Field-Collected Image Datasets

a: Papers using Private Field-Collected Apple Leaf Image Datasets

Li et al. [132] use the RegNet [133] architecture in their work to train on a dataset that was collected by the authors. The dataset, which contains 5 classes, was taken in field and totals 3,131 images after augmentation methods (rotation, translation, brightness, scaling) were applied to deal with the imbalance of the original dataset. This dataset is then split into train, validation and test sets 7:2:1. This data was then used to train RegNet, as well as ShuffleNet [63], MobileNetV3 [60], and EfficientNet-B0 [14] for comparison. Models were trained once with all pre-trained layers frozen and once with them all unfrozen to fine-tune them. The authors note that unfreezing all layers generated better results. Each of the models was also trained with 4 different optimizers, those being SGD, Adam, RAdam, and Ranger. These experiments showed that the RegNet model was the strongest among the ones tested and that the Adam optimizer produced the best results. Then RegNet with Adam was trained once more, with different learning rates (0.0001, 0.0005, 0.001, 0.005, 0.01, 0.05). The best performing model was trained with a learning rate of 0.0001, which reached 99.8% on the validation data and 99.23% on the testing set.

Shafik et al. [134] introduced an enhanced deep learning approach combining CNNs and LSTM networks, integrated with a majority voting ensemble classifier, for accurate detection and classification of plant diseases and pests. Deep features were extracted from pre-trained CNN models, including DenseNet201, GoogleNet, AlexNet, ResNet101, Xception, and ResNet50, and these features were passed through LSTM layers to build a robust detection framework. Their experiments were conducted using a dataset collected in Turkey comprising of 4,447 images of apple pests and diseases, divided into 15 classes, with image dimensions of 224×224 and 227×227 . To achieve optimal performance, key model configurations were adjusted, including varying the number of epochs in steps of 10 (ranging from 10 to 100) and fine-

tuning batch sizes. The proposed model outperformed other state-of-the-art methods, achieving an accuracy of 99.2%, validated against multiple classifiers, including LR, ELM, and LSTM. Pre-trained models such as VGG19 and AlexNet demonstrated improved accuracy when features extracted from the fc6 layer were used.

b: Papers using Private Field-Collected Cotton Leaf Image Datasets

Khan et al. [135] focused on identifying diseases in cotton leaves. Cotton leaves are affected by various diseases. The authors created a dataset of 2,385 images of healthy and diseased cotton leaves, collected from fields, and enhanced the dataset using data augmentation. They tested several models, including a Custom CNN, VGG16, and ResNet50, but their proposed meta deep learning model achieved the highest accuracy of 98.53%.

c: Papers using Private Field-Collected Rice Leaf Image Datasets

Nayak et al. [136] study the early detection of rice plant diseases. They used 2,259 images, along with 250 real-time validation images, to classify 12 rice diseases and nutrient deficiencies. Image segmentation techniques, such as foreground extraction, were applied to isolate affected areas of the plants. To make the solution practical, they optimized the models to work offline on smartphones and developed an application. The study compared various deep learning models, with DenseNet201, Xception, MobileNetV2, and ResNet50 achieving validation accuracies of 98.03%, 97.78%, 97.56%, and 97.18%, respectively. For cloud-based architectures, ResNet50 performed the best, while MobileNetV2 was ideal for smartphone applications due to its efficiency. An Android app, "Rice Disease Detector," integrated with the MobileNetV2 model was part of the paper.

Dogra et al. [137] propose a CNN-VGG19 model with transfer learning for the classification of rice leaf diseases. The dataset was collected by the authors themselves in field in India and includes 2,902 images. The proposed model managed to achieve 93.0% accuracy. Other metrics include sensitivity (89.9%), specificity (94.7%), precision (92.4%), and F1-score (90.5%), outperforming baseline models like CNN, VGG16, DNN, and RNN. The model processes images resized to 224×224 pixels, employing normalization and HSV-based augmentation. The training utilized a batch size of 200 and an optimized learning rate.

Ritharson et al. [138] focused on utilizing DL and TL techniques for rice leaf disease classification. A dataset of 5,932 self-generated images was used in the work. These images were categorized into 9 classes, covering healthy leaves and various degrees of mild and severe diseases. Data augmentation techniques, including rotation, cropping, and noise addition, were employed to improve model generalization. They used several models, including VGG16, ResNet50, DenseNet121, Xception, InceptionV3, and InceptionResNetV2, to assess their performance on the rice leaf

disease dataset. A custom-tailored VGG16 model, optimized with a learning rate between $1e^{-5}$ to $1e^{-2}$, dropout rates ranging from 0.2 to 0.4, and using optimizers like Adam, RMSprop, and SGD, demonstrated the best results. Hyperparameter tuning also incorporated weight decay for regularization and varied the number of training epochs between 10 and 100. The proposed custom VGG16 model achieved an accuracy of 99.7% on the 5,932-image dataset within 50 training epochs, surpassing several benchmarks. It outperformed Faster-RCNN, which achieved 98.85% accuracy on 2,400 images, and ResNet197, which reached 99.58% accuracy on 154,500 images but required 1,000 epochs.

d: Papers using Private Field-Collected Tomato Leaf Image Datasets

Sreedevi et al. [139] propose a tomato leaf disease classification method using CNN. The process begins with contrast enhancement for image pre-processing with GAN-based data augmentation system, followed by handling low-resolution problems through DCNN (ResNet, MobileNet, DenseNet, and Mask-RCNN). Disease segmentation is performed using Mask Region-Based Convolutional Neural Networks (Mask R-CNN). To optimize classifier scores, the study uses a Controlling Parameter-based Artificial Gorilla Troops Optimization (CP-AGTO) algorithm. The proposed method achieves an accuracy of 92% and a sensitivity of 93%. Its performance was evaluated using various metrics, demonstrating its effectiveness compared to recent approaches.

e: Papers using Private Field-Collected Soybean Leaf Image Datasets

Wang et al. [140] introduced a sliding segmentation method to segment images into 230×230 pixels using a sliding window technique. A CNN-based image recognition method was employed to classify soybean leaves at various disease stages. Transfer learning using classical deep learning networks was applied to leverage features from diseased and healthy leaves. The model optimization focused on tuning the optimization algorithm (SGD), epochs (200), and batch size (8) to enhance recognition accuracy. The Swin Transformer emerged as the best model, achieving an accuracy of 99.64% in identifying disease stages. This model enables precise identification of soybean leaf diseases, prevents pesticide misuse, and reduces yield losses.

Hang et al. [141] used Xception, MobileNetV2, NASNet-Mobile, and MobileNet (with ConvLSTM and squeeze and excitation (SE) layers added to the end), as well as a Fuzzy Ensemble of the models on a soybean dataset. The dataset was collected by the authors in China, in field, and contains 2,820 images across 7 classes. To extend the dataset, the authors opted to generate synthetic images via CycleGAN, which increased the total number of images to 8,456 images, roughly 1,200 images per class. The extended dataset was then split 6-1-3 (train, validation, test) before being fed to the pre-trained networks and the ensemble net. A learning rate of 0.0002, a batch size of 32, and the categorical cross

entropy loss function were used. The test accuracy evaluation reveals that the ensemble method reached the highest scores with 94.27%. Xception manages 91%, MobileNetV2 91.21%, NASNetMobile 90.39% and MobileNet 80.57% (all models with LSTM and SE added to the last layers).

f: Papers using Private Field-Collected Corn/Maize Leaf Image Datasets

Francisco et al. [142] suggested a corn plant disease severity classification system utilizing infrared thermography combined with DL techniques. A novel dataset of 2,279 corn thermograms was collected from the Institute of Plant Breeding in Los Baños, Philippines, categorizing infection severity into four levels: Highly Resistant (0%-5%), Moderately Resistant (10%-20%), Moderately Susceptible (30%-50%), and Susceptible (60%-100%). Three customized CNN architectures used and ensemble learning were developed and evaluated: Model 1 (two-stage binary classification using NASNetMobile and MobileNetV2), model 2 (multi-classification with EfficientNet-B1 and NASNetMobile), and finally, model C (integrating Model B with binary classification using MobileNetV2). They found that TL enhanced the accuracy of these models during training, with NASNetMobile achieving 93.8% for binary and 90.7% for multiclass classification, MobileNetV2 reaching 95.8% and 96.9% for binary classification, and EfficientNet-B1 attaining 91.1% for multiclass classification. In field testing, models 1, 2, and 3 achieved accuracies of 56.92%, 63.08%, and 76.92%, respectively.

g: Papers using Private Field-Collected Mulberry Leaf Image Datasets

Salam et al. [143], as part of the development of an Android app, tested several transfer-learning CNN on a mulberry dataset. This dataset was collected by the authors and after augmenting it with rotating, flipping, transformations, translation, and zooming it reached 6,000 images in 3 classes. The models (MobileNetV3Small, ResNet50, and VGG19) were slightly modified by appending 4 more convolutional layers to the back of them. Training was carried out with 5-fold cross-validation, with results being averaged over those runs. ResNet50 managed 94.4%, VGG19 achieved 90.89% and MobileNetV3-Small managed 92.5%.

h: Papers using Private Field-Collected Grape Leaf Image Datasets

Shah et al. [144] collected a dataset of grape leaves in India. The original dataset contained only 300 images over 4 classes, so the authors applied transformation, cropping, rescaling, resizing, rotation, brightness, contrast, and sharpness augmentations to inflate the dataset to 1,600 total images. The dataset was resized to 224×224 and split 60-20-20. VGG16, VGG19, InceptionV3, InceptionRes-NetV2, Xception, and ResNet50V2 were utilized by the authors in this study. When comparing the models results on the validation set, the results indicate that the ResNet50V2 model managed the highest accuracy with 98%. The other results were

VGG16 with 95%, VGG19 with 96%, InceptionV3 with 94%, InceptionRes-NetV2 with 94%, and Xception with 96%.

i: Papers using Private Field-Collected Chili Leaf Image Datasets

Winiarti et al. [145] compared the performance of a basic CNN, MobileNetV2, and VGG16 on a chili dataset that was comprised of both images from Kaggle and images collected by the authors. In total the dataset comes up to 250 images across 5 classes (yellowish, white fly, leaf spot, curl leaf, healthy). Training results show that VGG managed the highest score with 94%, with MobileNetV2 coming in second at 91%, with the CNN only reached 88%.

j: Papers using Private Field-Collected Mango Leaf Image Datasets

Chen et al. [146] utilized Class Activation Mapping (CAM) heatmaps to confirm the ability to focus on disease-relevant regions in the images. They focused on the detection of mango leaf diseases using a Multi-Scale and Multi-Pooling Convolutional Neural Network (MSMP-CNN) model. The dataset comprises of 3,776 images divided into six classes. To enhance data variability, augmentation techniques such as horizontal flipping were applied. Transfer Learning and fine-tuning were subsequently employed to enhance classification performance. Initially, the MSMP-CNN model achieved an accuracy of 95%, which increased to 98.5% following the implementation of TL and fine-tuning.

3) Papers using Hybrid Condition Image Datasets

a: Papers using Private Hybrid Condition Rice Leaf Image Datasets

Ahad et al. [147] address limitations in rice disease detection research by comparing the performance of six CNN-based architectures (DenseNet121, Inceptionv3, MobileNetV2, ResNext101, ResNet152V, and SeresNext101) using a dataset of 9 prevalent rice diseases in Bangladesh. This research evaluates individual CNN networks, transfer learning, and ensemble techniques to enhance detection and localization accuracy. The ensemble model, DEX (combining DenseNet121, EfficientNet-B7, and Xception), achieved the highest accuracy of 98%, outperforming individual CNN architectures. Additionally, applying transfer learning improved accuracy by 17% for SeresNext101, demonstrating its effectiveness in rice disease detection. The models were optimized with 175 epochs, a learning rate of 0.0001, $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 1 \times 10^{-7}$.

b: Papers using Private Hybrid Condition Tomato Leaf Image Datasets

Santoso et al. [148] focused on comparing the performance of two deep learning models, a CNN and DenseNet121. The research utilized a combined dataset that included field data collected from tomato farms in Central Java, Indonesia, and the publicly available PlantVillage dataset, in a total of

18,815 images representing 13 different disease categories. To prepare the data, images were normalized to 400x400 pixels, and techniques such as data augmentation and class balancing were applied to improve the dataset's quality before training and testing the models. Finally images were resized to 224x224x3. The CNN model achieved a training accuracy of 90.41% and a validation accuracy of 83.33%. The model was initialized with pre-trained ImageNet weights and trained using the categorical cross-entropy loss function and the SGD optimizer with a learning rate of 0.001. DenseNet121 demonstrated superior performance, achieving a training accuracy of 98.27%, a validation accuracy of 87.47%, and average precision, recall, and F1-scores of 88%, 87%, and 87%.

c: Papers using Private Lab-Based Banana Leaf Image Datasets

Genet et al. [149] present a CNN-based model for detecting banana plant diseases, focusing on 3 classes: healthy, xanthomonas wilt, and sigatoka leaf spot. A dataset of 1,982 banana leaf images, resized to 150x150 pixels, was used for training. Techniques such as data augmentation and transfer learning were applied using various architectures, including MobileNet, EfficientNet, VGG16, VGG19, and InceptionV3. The model was trained for 30 epochs with RMSprop optimizer, a learning rate of 0.0001, and a dropout rate of 0.5. Among the tested models, VGG16 achieved the highest classification accuracy of 81.53%, proving its potential for disease detection in banana.

C. REAL WORLD USE-CASES

Reda et al. [77] developed AgroAId, a mobile application for real-time plant disease identification using a TensorFlow-based DL model. The app allows users to classify plant images by selecting "Classify a New Plant", which leads to an image selection activity. Users can input images either by capturing a photo via the device's camera or uploading from the gallery. Once an image is selected and confirmed, the model processes the image in the background. The classification results are displayed on a dedicated screen, showing the predicted class, confidence score, image, date/time, and location of classification. These results are stored in the user's history, and analytics are updated with class, region, and season details. Users can also access plant-care information related to the predicted combination. The model integrated in the app was verified using images from the PlantVillage dataset, confirming its effectiveness in accurate disease classification.

Nayak et al. [136] exhibited a smartphone application for rice disease classification and nutrient deficiency detection. The application captures images directly from the field using the smartphone's camera. Once an image is selected, pre-processing is performed in the background. The app integrates the MobileNetV2 model, which extracts features from the image using pretrained weights. The model is validated with the input image, and its parameters are saved in memory for future training with new data. Additionally, the image

data is stored on a cloud server. After confirming the selected image and completing the feature extraction process, the system proceeds to the prediction stage. Finally, the app displays the identified disease name along with the detected nutrient deficiency related to the plant.

Salam *et al.* [143] built a lightweight Android app for real-time plant disease detection using a quantized TFLite model. The trained CNN was converted to 8-bit TFLite format, reducing model size and power consumption while improving inference speed. An API was used to normalize the input data. The app allows users to upload or capture leaf images, which are pre-processed before being passed to the model and classified into two disease categories and healthy. Initially, the model was stored locally within the app. The app was tested using both laboratory images and real field images captured directly from mulberry farms. It achieved inference times of 4–12 ms with high accuracy on a Poco X2 smartphone, demonstrating fast, on-device classification. The combination of quantization, transfer learning, and optional GPU-accelerated inference enables efficient and reliable disease detection in mobile environments.

All papers with a compassion of at least 4 TL papers will be listed in Tables 4, 5, 6, 7, 8, 9, 10, and 11. Each paper will be listed with the TL model results and the different metrics that each paper provides in their result tables, as well as the dataset that is used and the relevant reference to the dataset and the paper that provides the results.

IV. DISCUSSION

After having reviewed 84 papers in the previous section, this section will analyze these papers and present recent best practices and trends that can be found in these works. We will present the most popular models, datasets, parameters, and more in this section, starting with the models.

A. MODELS

When looking at Tables 12 and 13 and Figure 10, one can see that while there is a wide variety of transfer learning CNN models used in recent papers, there also seem to be a number of tendencies that become apparent. Although the VGG models were some of the first popular deep CNN foundational models, they still see a really large number of use. Among the papers reviewed in this work, VGG16 is the most used model, with VGG19 being the 7th most popular model. The VGG models, when compared to other CNN architecture families, come in 3rd among the most used models. This showcases that VGG can still generate models that are competitive in certain cases, while being highly available, not too heavy, and also more easy to understand, maybe especially to researchers that focus more on the plant pathology side of thing and are not too deeply involved in the field of DL. Some other models like Inception (who's family is the second most used) based InceptionV3 is the second most popular model, not far behind VGG16, with MobileNetV2 of the MobileNet family (4th most popular architecture family) coming in third. ResNet50 (ResNet family; most popular) also saw a lot of work, as did

the DenseNet and EfficientNet families. This proves that a wide range of different models are being used to train plant leaf disease classification CNN models using TL. While there are definite tendencies towards certain architectures, there seems to not be a concrete consensus among researchers. Also noteworthy is, that newer models, like ConvNeXt for example, have yet to see much attention in this field, with more established and somewhat older models still being much more popular.

These usage statistic do, however not mean, that these often used models are automatically the strongest or best suited. To analyze the model performance across different studies, we will rank models according to their quartiles. Papers that compare 4 or more TL models are considered (to assure proper comparability). Then, for each study, models are ranked in quartiles based on their accuracy score (1st quartile (top 25% performance in the respective paper), 2nd quartile (top 50%), 3rd quartile (top 75%), or the 4th quartile), since accuracy is used in all but one study ([111]), where F1-Score will be used instead. Then only models that are used at least 3 times are included in the average rankings (to provide some robustness and generalization). Quartiles are used since different studies use different models and also different amounts of models on different datasets and in different setups and scenarios, making a direct ranking or a average accuracy score faulty and unfair. The model's quartiles ranking will be averaged across all eligible studies once for all papers using field data, once for all papers using lab data, once for all papers using hybrid data, once for all papers using PlantVillage specifically (due to its wide spread usage), and once for all datasets. This is done in an effort to identify not only the most used models, as was discussed above, but also the stronger models among them, specifically for this field, for each dataset condition, for models with enough usage and data.

On the field datasets, the Xception model came out on top with a average quartile rating of 1.167, with only 1st quartile results except for one in the 2nd. Following Xception are DenseNet 169, EfficientNet-B0, and DenseNet121, all with average scores of under 2. On the other end, models like InceptionV3, ResNet50V2, VGG19, and InceptionResNetV2, all have average quartile score of over 3. It can be observed that DenseNet family models all perform with an average score of 2 or better, while no ResNet model managed to perform better than 2. MobileNet models, as well as the VGG16 model are in the midfield of average scores, while being smaller than many other models in terms of parameter counts, offering up a more lightweight alternative with decent scores.

On the lab datasets the DenseNet201 architecture managed to reach first place, with EfficientNet-B0 reached second place, as it did on the field data. InceptionV3 seems to be better suited to handle lab data, with its results being better than they were on the field datasets. NASNetMobile again performed near the bottom, coming in last on lab data with a score of 3.75, after only managing 2.667 on field data.

When looking at PlantVillage data results, DenseNet and

TABLE 4. Comparative table of different models on field datasets. Refer to Section III or the paper itself for detailed information about the methodology used to achieve these score, since each paper uses slightly different approaches to reach the scores reported here. The metrics usage in this table depends on the metrics provided in any given paper. This table includes papers that use at least 4 different properly cited transfer-learned models on the same dataset with the same conditions and well defined and provided metrics and results, to allow for actual comparison between them.

Dataset	Model(s)	Metrics				
Own Apple Dataset (Field) [132]	Multiple	SGD	Adam	RAdam	Ranger	(All Acc.)
	ShuffleNet [132]	62.3%	90.2%	77.9%	72.1%	
	MobileNetV3 [132]	89.9%	93.4%	93.1%	93.2%	
	EfficientNet-B0 [132]	86.4%	94.3%	94.0%	93.4%	
	RegNet [132]	98.2%	99.8%	99.8%	99.9%	
DiaMOS [29]	Multiple	Precision	Recall	F1-Score	Accuracy	
	ResNet50 [105]	47.00%	38.53%	40.54%	69.90%	
	ResNet50-CBAM [105]	48.24%	39.06%	41.08%	71.36%	
	VGG19 [105]	64.45%	64.85%	64.43%	73.09%	
	VGG19-CBAM [105]	68.10%	64.76%	65.97%	73.42%	
	InceptionV3 [105]	62.60%	61.71%	61.96%	76.61%	
	InceptionV3-CBAM [105]	78.13%	65.45%	66.73%	77.87%	
	MobileNetV2 [105]	66.47%	66.54%	66.49%	82.06%	
	MobileNetV2-CBAM [105]	70.68%	70.00%	70.27%	83.99%	
	EfficientNet-B0 [105]	67.38%	67.82%	67.59%	85.82%	
Cassava [30]	Multiple	Accuracy	F1-Score			
	VGG16 [110]	60.63%	40%			
	VGG19 [110]	62.90%	43%			
	ResNet50 [110]	73.46%	55%			
	ResNet50V2 [110]	72.34%	53%			
	ResNet101V2 [110]	71.64%	52%			
	ResNet152V2 [110]	70.14%	45%			
	InceptionV3 [110]	69.91%	47%			
	InceptionResNetV2 [110]	69.67%	46%			
	MobileNetV2 [110]	71.17%	55%			
	DenseNet121 [110]	73.27%	53%			
	DenseNet169 [110]	74.77%	59%			
	DenseNet201 [110]	70.84%	52%			
Rice Disease Dataset DMD [31]	Multiple	Accuracy	Precision	Recall	F1-Score	Specificity
	VGG16 [108]	87.47%	89%	89%	88%	80.82%
	VGG19 [108]	84.84%	85%	85%	85%	79.45%
	ResNet50 [108]	74.66%	75%	75%	74%	76.78%
	ResNet101 [108]	70.32%	71%	72%	71%	75.75%
	NASNetMobile [108]	91.01%	92%	91%	91%	98.24%
	InceptionV3 [108]	91.93%	93%	93%	93%	86.95%
	Xception [108]	92.01%	92%	92%	92%	83.09%
	DenseNet121 [108]	93.87%	94%	94%	94%	83.78%
	MobileNet [108]	90.96%	92%	91%	91%	83.33%
	InceptionResNet [108]	93.87%	95%	95%	94%	89.85%
Own [134] (field)	Multiple	Accuracy	F1-Score	Sensitivity	Specificity	
	ResNet50 [134]	95.38%	93.28%	94.91%	99.36%	
	Xception [134]	91.28%	88.33%	89.22%	98.70%	
	InceptionResNetV2 [134]	89.23%	85.15%	85.09%	98.48%	
	ResNet101 [134]	96.41%	95.94%	95.81%	99.43%	
	GoogleNet [134]	95.30%	93.53%	92.29%	95.75%	
	InceptionV3 [134]	92.31%	88.76%	87.05%	98.90%	
	VGG19 [134]	94.87%	93.87%	92.24%	99.14%	
	VGG16 [134]	96.92%	93.87%	92.24%	99.14%	
	AlexNet [134]	93.33%	88.30%	88.02%	99.00%	
Rice Dataset (field))	Multiple	Accuracy				
	AlexNet [109]	69.00%				
	VGG16 [109]	86.98%				
	VGG19 [109]	79.55%				
	Resnet50 [109]	60.50%				
	InceptionV3 [109]	78.81%				
	Xception [109]	87.36%				
	MobileNet [109]	83.64%				
	DenseNet201 [109]	82.89%				

TABLE 5. Continuation of Table 4.

Dataset	Model(s)	Metrics			
Own Soybean [147] (field)	Multiple	Accuracy			
	Xception [141]	89.04%			
	MobileNetV2 [141]	88.04%			
	NASNetMobile [141]	88.54%			
	MobileNet [141]	78.66%			
Own Rice Dataset [138] (field)	Multiple	Precision	Recall	F1-Score	Accuracy
	Xception [138]	97.7%	97.7%	97.7%	97.7%
	DenseNet121 [138]	93.8%	93.8%	93.8%	93.8%
	InceptionResNetV2 [138]	93.3%	93.3%	93.3%	93.3%
	ResNet50 [138]	81.1%	81.1%	81.1%	81.1%
	InceptionV3 [138]	95.5%	95.5%	95.5%	95.5%
Own Grape Dataset [144] (field)	VGG16 [138]	98.8%	98.8%	98.8%	98.8%
	Multiple	Accuracy			
	InceptionV3 [144]	94%			
	VGG16 [144]	95%			
	VGG19 [144]	96%			
	ResNet50V2 [144]	98%			
	InceptionResNetV2 [144]	94%			
Banana [38] + Black Gram [39]	Xception [144]	96%			
	Multiple	Accuracy	Precision	Recall	
	MobileNetV2 [114]	75.31%	76.60%	74.07%	
	DenseNet121 [114]	81.48%	81.74%	81.07%	
	ResNet152V2 [114]	79.42%	80.42%	79.42%	
	DenseNet169 [114]	81.07%	81.33%	80.66%	
	DenseNet201 [114]	82.30%	82.57%	81.89%	
	InceptionV3 [114]	74.07%	75.86%	72.43%	
	NASNetLarge [114]	78.60%	79.25%	78.60%	
	InceptionResNetV2 [114]	69.55%	72.94%	65.43%	
Tomato leaf disease detection [41]	EfficientNetV2S [114]	78.19%	79.08%	77.78%	
	EfficientNetV2L [114]	74.07%	74.69%	74.07%	
	Multiple	Accuracy	Sensitivity	F1-Score	
	DenseNet201 [116]	97.56%	97.45%	97.46%	
	MobileNet [116]	90.89%	90.61%	90.52%	
	LCNet [116]	89.67%	89.26%	89.23%	
Date Palm data [42]	GhostNet [116]	90.78%	90.50%	90.37%	
	CoAtNet [116]	97.67%	97.57%	97.59%	
	ConvMixer [116]	80.56%	79.79%	79.83%	
	FasterNet [116]	93.22%	92.91%	92.92%	
	Multiple	Accuracy			
	ResNet152V2 [117]	98.47%			
	VGG19 [117]	91.25%			
	DenseNet201 [117]	98.89%			
	MobileNetV2 [117]	99.72%			
	InceptionResNetV2 [117]	98.19%			

EfficientNet again end up at the top, with DenseNet121 scoring 1.2 and EfficientNet-B0 reaching 1.5. MobileNet too reached 1.5, showing good performance on hybrid data, with DenseNet201 also scoring with better than 2 results at 1.75. Xception, which managed the best results on pure field data, came in second last on PlantVillage data, suggesting that it can succeed in field cases (or in cases with lass classes and less images) and that it struggles in the PlantVillage case with more uniform data (and many images and classes).

Hybrid data results have MobileNet on top with an average score of 1.333. Xception again managed to reach good scores with an average of 2 and even InceptionV3, which struggled in other scenarios achieved good results on the hybrid datasets. Both ResNet models were ranked in at the very

bottom with scores of only 3.5, suggesting that they are less suitable in hybrid cases.

When looking at the overall results, it becomes immediately apparent that EfficientNet models and DenseNet models were clearly the best model families among the models used in the scope of this review. The top 7 models are all part of these two model families, all managing scores better than 2. The only other model that managed to achieve scores better than 2 was Xception, which ranked in at 8th and did well in field and hybrid cases. One could also see that most ResNet models were ranking in among the middle of the tested models. At the bottom older models like AlexNet, VGG19, SqueezeNet can be found. Additionally NASNet models did not fare well, ranking near the bottom in nearly all categories.

TABLE 6. Continuation of Tables 4 and 5.

Dataset	Model(s)	Metrics		
iCassava [30]	Multiple	Precision	Recall	F1-Score
	DenseNet121 [111]	67%	53%	53%
	DenseNet169 [111]	66%	63%	63%
	DenseNet201 [111]	66%	64%	62%
	EfficientNet-B0 [111]	68%	68%	67%
	EfficientNet-B1 [111]	72%	63%	63%
	EfficientNet-B2 [111]	71%	69%	69%
	EfficientNet-B3 [111]	70%	68%	68%
	EfficientNet-B4 [111]	73%	69%	70%
	EfficientNet-B5 [111]	72%	69%	69%
	EfficientNet-B6 [111]	72%	69%	69%
	EfficientNet-B7 [111]	72%	69%	69%
	InceptionResNetV2 [111]	68%	69%	67%
	NASNetLarge [111]	04%	20%	07%
	NASNetMobile [111]	05%	20%	08%
	ResNet50 [111]	62%	59%	59%
	ResNet50V2 [111]	56%	51%	48%
	ResNet101 [111]	08%	21%	06%
	ResNet101V2 [111]	52%	47%	44%
	ResNet152 [111]	63%	54%	52%
	ResNet152V2 [111]	55%	46%	43%
	VGG16 [111]	35%	32%	27%
	VGG19 [111]	50%	43%	43%
	Xception [111]	70%	69%	69%
	A_EfficientNetB0 [111]	75%	74%	74%
	A_EfficientNetB1 [111]	76%	72%	71%
	A_EfficientNetB2 [111]	76%	72%	73%
	A_EfficientNetB3 [111]	77%	76%	74%
	A_EfficientNetB4 [111]	79%	78%	77%
	A_EfficientNetB5 [111]	81%	75%	76%
	A_EfficientNetB6 [111]	81%	75%	76%
	A_EfficientNetB7 [111]	79%	68%	70%
	A_M_EfficientNet [111]	88%	86%	87%
	A_L_EfficientNet [111]	86%	84%	85%

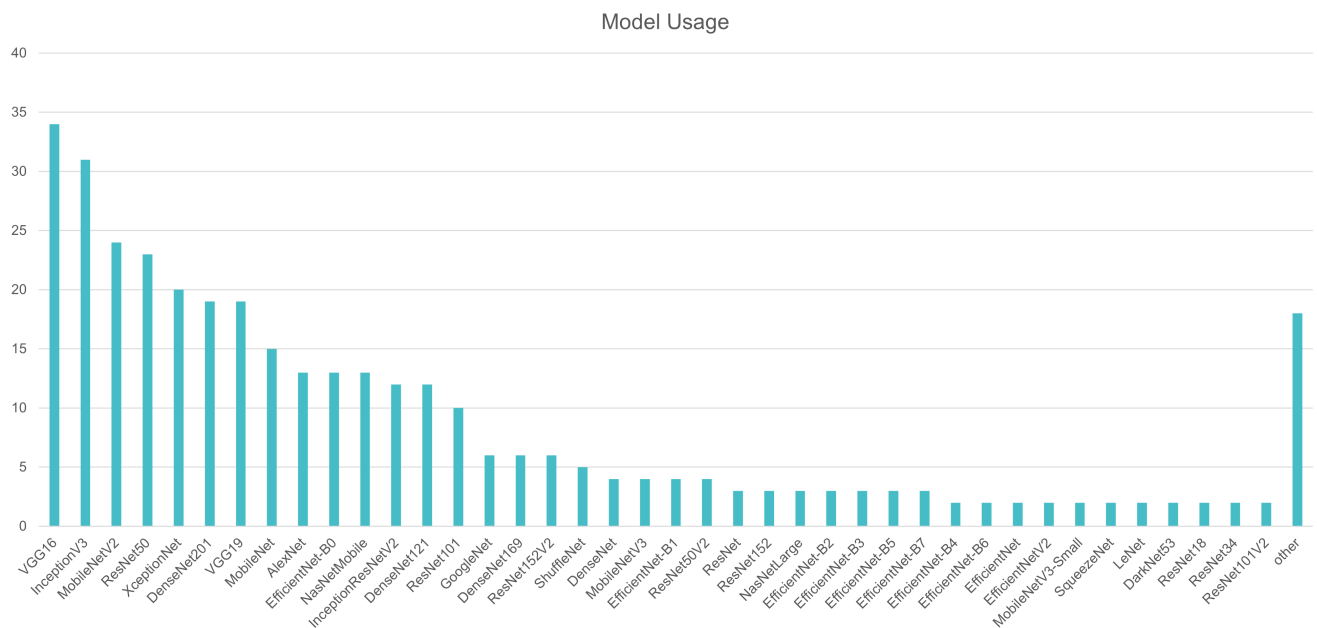


FIGURE 10. Figure depicting the ratio of different models used.

TABLE 7. Comparative table of different models on lab datasets. Refer to Section III or the paper itself for detailed information about the methodology used to achieve these score, since each paper uses slightly different approaches to reach the scores reported here. The metrics usage in this table depends on the metrics provided in any given paper. This table includes papers that use at least 4 different properly cited transfer-learned models on the same dataset with the same conditions and well defined and provided metrics and results, to allow for actual comparison between them.

Dataset	Model(s)	Metrics				
Own Chili Dataset (Field) [131]	Multiple	Accuracy	Precision	Recall	F1-Score	Specificity
	AlexNet [131]	81.47%	81.55%	81.47%	81.42%	96.29%
	DarkNet53 [131]	82.95%	83.23%	82.95%	82.95%	96.58%
	DenseNet201 [131]	83.37%	83.50%	83.37%	83.20%	96.67%
	EfficientNet-B0 [131]	81.28%	81.68%	81.28%	81.25%	96.25%
	InceptionV3 [131]	82.12%	82.97%	82.12%	82.16%	96.42%
	MobileNetv2 [131]	79.40%	79.97%	79.40%	79.42%	95.88%
	NasNetLarge [131]	80.45%	80.95%	80.45%	80.36%	96.08%
	ResNet101 [131]	81.70%	81.73%	81.70%	81.61%	96.33%
	ShuffleNet [131]	79.42%	79.90%	79.42%	79.34%	95.88%
	SqueezeNet [131]	81.68%	81.88%	81.68%	81.71%	96.33%
	Xception [131]	81.46%	81.57%	81.46%	81.41%	96.29%
	VGG19 [131]	83.54%	84.46%	83.54%	83.72%	96.71%
Rice Leaf 5 Disease [26]	Multiple	Accuracy	Precision	Recall	Specificity	F1-Score
	GoogleNet [95]	83.87%	83.73%	84.04%	96.77%	83.79%
	VGG16 [95]	88.71%	88.85%	88.89%	97.74%	88.35%
	VGG19 [95]	87.10%	86.74%	87.25%	97.42%	86.81%
	DenseNet201 [95]	89.86%	90.05%	90.05%	97.97%	89.86%
	AlexNet [95]	86.18%	87.64%	86.29%	97.22%	85.54%
BRACOL [28]	Multiple	Precision	Sensitivity	Specificity	F1-Score	Accuracy
	VGG16 [99]	94.4%	94.0%	98.6%	94.1%	94.2%
	InceptionV3 [99]	83.5%	85.1%	96.3%	83.5%	83.9%
	ResNet152 [99]	94.0%	93.2%	98.5%	93.3%	93.8%
	Xception [99]	85.5%	85.3%	96.6%	85.2%	85.4%
	MobileNetV2 [99]	76.8%	74.1%	94.5%	73.5%	74.6%
	DenseNet201 [99]	84.7%	83.2%	96.3%	83.3%	83.8%
	InceptionResNetV2 [99]	86.7%	86.0%	97.0%	86.1%	86.9%
	NASNetMobile [99]	85.1%	83.1%	96.3%	83.3%	83.8%
	EfficientNet-B0 [99]	95.2%	94.8%	98.8%	94.9%	95.0%
RLDIS [24] + RLDD [32]	Multiple	Accuracy	Precision	Recall	F1-Score	Specificity
	InceptionV3 [96]	99.64%	98.23%	98.21%	98.20%	99.80%
	InceptionResNetV2 [96]	99.62%	98.14%	98.12%	98.12%	99.79%
	DenseNet201 [96]	99.61%	98.13%	98.08%	98.08%	99.78%
	Xception [96]	99.61%	98.12%	98.06%	98.07%	99.78%
	EfficientNet-B0 [96]	99.56%	97.93%	97.80%	97.82%	99.75%
	ResNet101 [96]	99.55%	97.77%	97.74%	97.73%	99.75%
	MobileNetV2 [96]	99.53%	97.78%	97.68%	97.68%	99.74%
	NASNetMobile [96]	99.48%	97.43%	97.40%	97.40%	99.71%
	ResNet50 [96]	99.46%	97.44%	97.33%	97.32%	99.70%
	ShuffleNet [96]	99.41%	97.13%	97.05%	97.05%	99.67%
	GoogleNet [96]	99.40%	97.12%	96.99%	97.00%	99.66%
	DarkNet53 [96]	99.35%	96.89%	96.75%	96.73%	98.31%
	SqueezNet [96]	99.09%	95.70%	95.45%	95.46%	99.49%
	VGG16 [96]	98.47%	92.59%	92.31%	92.30%	99.15%
	AlexNet [96]	97.35%	87.74%	86.73%	86.50%	98.53%
	Various Sources (lab)	Multiple	Accuracy	F1-Score		
EfficientNet-B5 [84]		99.20%	99.20%			
InceptionV3 [84]		58.70%	58.50%			
MobileNet [84]		82.00%	82.00%			
ResNet50 [84]		98.98%	98.98%			
VGG16 [84]		98.29%	98.29%			
Potato Leaf Disease Dataset [25]	Multiple	Accuracy				
	NASNetMobile [101]	95.33%				
	VGG19 [101]	95.67%				
	DensNet169 [101]	99.67%				
	ResNet50V2 [101]	97.00%				
	InceptionV3 [101]	97.00%				
	Xception [101]	96.00%				

TABLE 8. Continuation of Table 7.

Dataset	Model(s)	Metrics				
Wheat Disease Detection [34]	Multiple	Accuracy	Sensitivity	F1-Score		
	EfficientNet-B2 [104]	94.58%	94.66%	94.58%		
	MobileNet [104]	95.97%	95.97%	95.95%		
	Xception [104]	98.06%	98.02%	98.03%		
	NASNetMobile [104]	48.61%	47.61%	38.70%		
	InceptionV3 [104]	97.92%	97.88%	97.89%		
	DenseNet121 [104]	97.78%	97.74%	97.77%		
	DenseNet169 [104]	97.64%	97.60%	97.61%		
	DenseNet201 [104]	97.92%	97.88%	97.90%		
	RegNetY080 [104]	97.64%	97.64%	97.64%		
	ResNet50V2 [104]	97.22%	97.17%	97.19%		
	ResNet101V2 [104]	97.92%	97.88%	97.89%		
	ResNetRS50 [104]	97.36%	97.32%	97.35%		
	ResNetRS101 [104]	96.53%	96.53%	96.51%		
Own [76] (Grape) + AI Challenger 2018	Multiple	Accuracy	Recall	Precision	F1-Score	Specificity
	VGG16 [76]	92.12%	93.12%	93.96%	93.08%	98.01%
	ResNet50 [76]	94.86%	93.65%	94.21%	93.98%	99.02%
	SqueezeNet [76]	93.65%	92.35%	93.24%	92.68%	93.88%
	MobileNetV1 [76]	88.16%	79.31%	85.89%	78.25%	76.28%
	MobileNetV2 [76]	96.23%	94.39%	96.72%	92.58%	94.23%
	MobileNetV3 [76]	96.89%	95.03%	96.57%	96.81%	99.54%
	ShuffleNet [76]	95.47%	94.32%	95.17%	94.21%	99.14%
	Xception [76]	94.98%	94.63%	94.87%	95.24%	99.35%
	IresNet [76]	95.46%	94.93%	95.27%	94.62%	98.57%
	GC-MobileNet [76]	98.65%	98.59%	98.57%	98.97%	99.87%
	Multiple	Accuracy				
	VGG16 [92]	93.5%				
	Inception [92]	94.2%				
	ResNet [92]	95.1%				
	EfficientNetV2 [92]	95.2%				
	Proposed Approach [92]	97.2%				
RoseNet [150]	Multiple	Accuracy				
	VGG16 [92]	93.5%				
	Inception [92]	94.2%				
	ResNet [92]	95.1%				
	EfficientNetV2 [92]	95.2%				
	Proposed Approach [92]	97.2%				

B. DATASETS

When looking at the usage of datasets (Table 19 and Figure 11) across all the papers reviewed, some trends can also be identified. For one, it is immediately obvious that PlantVillage is by far the most used dataset. Not a single other dataset managed more than 3 appearances in the paper selected for review, although one should note that PlantVillage is often used in different configurations (as a whole, or in subsets). One one hand, this comes with the benefit of more comparative studies, since using the same datasets offers some comparatives across different studies. However, on the other hand, this is also somewhat of an issue, since PlantVillage is a lab dataset, models tend to always achieve very high results. This makes it much harder to actually compare the different results generated by different publications, since the differences are often rather small, because most of the models manage well above 90 or even 95%. Additionally, lab data trained models are less suitable for real life field conditions [17]–[20]. Models face a much bigger challenge with field datasets. Field datasets are more complex and less optimized for DL models, but much closer to the conditions found on real farms. Such datasets would therefore be preferable for benchmarking and comparing, as model performance could be better compared. However, a dataset of PlantVillage's size

and variety does not yet exist in field or hybrid conditions.

Most new datasets introduced in current research ("Own" in Table 19, where each of the 24 datasets are different datasets), which make up the second largest set of datasets, are, however, kept private (see Table 23), which limits comparability and also limits the public pool of available and usable data. One can also see that most datasets are field datasets (see Table 21, where the first value are overall statistics and numbers in parentheses are the values among "own" datasets), but none of the field datasets seem to have positioned themselves in a position similar to PlantVillage, probably due to PlantVillage's size and diversity, so most of the field datasets see little use comparatively. The origin of the different datasets is presented in Table 20. This is relevant because different plant are grown in different areas of the world, and even the same plants showcase different diseases in different locations, meaning that more diverse locations would also be able to provide more diverse data to train more robust models.

When taking a look at the average accuracy scores for each dataset type in Table 22, one can see that the average for PlantVillage is very high, with 94.41% average accuracy across 77 different models. This score is higher than that of any other dataset type, even that of other lab datasets, which

TABLE 9. Comparative table of different models on hybrid datasets. Refer to Section III or the paper itself for detailed information about the methodology used to achieve these score, since each paper uses slightly different approaches to reach the scores reported here. The metrics usage in this table depends on the metrics provided in any given paper. This table includes papers that use at least 4 different properly cited transfer-learned models on the same dataset with the same conditions and well defined and provided metrics and results, to allow for actual comparison between them.

Dataset	Model(s)	Metrics			
PV [16] + PD [17] + CD&S [27]	Multiple	Accuracy			
	AlexNet [118]	87.12%			
	VGG19 [118]	90.34 %			
	InceptionV3 [118]	92.82 %			
	ResNet50 [118]	87.58 %			
	DenseNet201 [118]	94.17%			
	MobileNetv2 [118]	84.38 %			
	EfficientNet [118]	93.52 %			
	EfficientNetv2 [118]	94.37 %			
Various Sources [66] (hybrid)	Multiple	Accuracy	Recall	Specificity	F1-Score
	Xception [125]	99.2%	94.6%	97.7%	98.8%
	InceptionV3 [125]	98.7%	93.4%	97.1%	98.4%
	DenseNet [125]	98.3%	92.1%	96.5%	97.4%
	AlexNet [125]	98.1%	90.9%	95.1%	96.3%
	ResNet [125]	98.5%	91.5%	92.5%	93.2%
Own [147] + RLDD [32] (hybrid)	Multiple	Accuracy			
	DenseNet121 [147]	97%			
	Inceptionv3 [147]	97%			
	MobileNetV2 [147]	94%			
	ResNext101 [147]	96%			
	ResNet152V2 [147]	93%			
	Seresnext101 [147]	79%			
RLDIS [24]	Multiple	Accuracy			
	DenseNet201 [123]	99.10%			
	EfficientNet-B3 [123]	100.00%			
	InceptionResNetV2 [123]	26.64%			
	MobileNetV2 [123]	100.00%			
	ResNet152V2 [123]	100.00%			
	VGG16 [123]	100.00%			
	VGG19 [123]	99.33%			
	VGG19 [123]	98.65%			
Various Sources (Maize) (hybrid)	Multiple	Accuracy			
	ResNet50 [126]	56.04%			
	ResNet101 [126]	51.80%			
	MobileNetV2 [126]	94.04%			
	MobileNet [126]	94.64%			
	InceptionV3 [126]	94.48%			
	InceptionResNetV2 [126]	93.12%			
	Xception [126]	95.80%			
Various Sources (Rice) (hybrid)	Multiple	Accuracy			
	ResNet50 [126]	62.32%			
	ResNet101 [126]	60.68%			
	MobileNetV2 [126]	95.84%			
	MobileNet [126]	95.72%			
	InceptionV3 [126]	96.20%			
	InceptionResNetV2 [126]	95.32%			
	Xception [126]	91.20%			
Various Sources (Wheat) (hybrid)	Multiple	Accuracy			
	ResNet50 [126]	55.36%			
	ResNet101 [126]	52.99%			
	MobileNetV2 [126]	96.32%			
	MobileNet [126]	96.28%			
	InceptionV3 [126]	95.08%			
	InceptionResNetV2 [126]	94.88%			
	Xception [126]	88.84%			
	VGG16 [126]	40.18%			

TABLE 10. Comparative table of different models on the PlantVillage datasets. Refer to Section III or the paper itself for detailed information about the methodology used to achieve these score, since each paper uses slightly different approaches to reach the scores reported here. The metrics usage in this table depends on the metrics provided in any given paper. This table includes papers that use at least 4 different properly cited transfer-learned models on the same dataset with the same conditions and well defined and provided metrics and results, to allow for actual comparison between them.

Dataset	Model(s)	Metrics			
PlantVillage [16] (Apple only)	Multiple	Accuracy			
	DenseNet169 [46]	97.52%			
	DenseNet201 [46]	98.75%			
	InceptionResNetV2 [46]	95.27%			
	InceptionV3 [46]	94.87%			
	MobileNet [46]	98.66%			
	MobileNetV2 [46]	98.29%			
	ResNet50 [46]	74.29%			
	VGG16 [46]	98.28%			
	VGG19 [46]	97.00%			
	Xception [46]	95.11%			
PlantVillage [16]	Multiple	Accuracy	F1-Score		
	MobileNetV2 [77]	77.375%	76.500%		
	NasNetMobile [77]	81.625%	81.000%		
	MobileNet [77]	83.750%	84.000%		
	EfficientNet-B0 [77]	99.000%	99.375%		
PlantVillage [16] (Maize only)	Multiple	Accuracy			
	InceptionV3 [47]	99%			
	DenseNet121 [47]	99%			
	DenseNet201 [47]	99%			
	MobileNet [47]	99%			
	VGG16 [47]	99%			
	EfficientNet-B0 [47]	99%			
	LeNet [47]	94%			
PlantVillage [16]	Multiple	F1-Score			
	EfficientNet-B0 [78]	99.6995%			
	EfficientNet-B1 [78]	99.6995%			
	EfficientNet-B2 [78]	99.5793%			
	EfficientNet-B3 [78]	99.7596%			
	EfficientNet-B4 [78]	99.6995%			
	EfficientNet-B5 [78]	99.7596%			
	EfficientNet-B6 [78]	99.8197%			
	EfficientNet-B7 [78]	99.8197%			
PlantVillage [16]	Multiple	Accuracy			
	InceptionV4 [79]	97.59%			
	VGG16 [79]	82.75%			
	ResNet50 [79]	98.73%			
	DenseNet121 [79]	99.81%			
PlantVillage [16] (23 classes)	Multiple	Accuracy	Precision	Recall	F1-Score
	InceptionV3 [80]	98.01%	96.52%	97.53%	96.52%
	ResNet50 [80]	93.57%	92.41%	91.64%	91.69%
	VGG16 [80]	87.50%	90.43%	90.17%	89.77%
	VGG19 [80]	86.70%	88.18%	86.94%	86.68%
PlantVillage [66] (Grape only)	Multiple	Adam	SGD	(All Acc.)	
	VGG16 [83]	97.91%	96.88%		
	ResNet50 [83]	82.64%	68.72%		
	InceptionV3 [83]	98.03%	98.28%		
	DenseNet121 [83]	98.40%	98.40%		

TABLE 11. Continuation of Table 10.

Dataset	Model(s)	Metrics			
PlantVillage [16] (Potato only)	Multiple	Accuracy			
	ResNet50 [102]	93.29%			
	VGG16 [102]	93.05%			
	Xception [102]	98.84%			
	InceptionV3 [102]	98.09%			
PlantVillage [16]	Multiple	Accuracy			
	DenseNet201 [87]	93.48%			
	ResNet101 [87]	93.25%			
	ResNet50 [87]	93.03%			
	ResNet18 [87]	91.45%			
	GoogleNet [87]	87.62%			
	AlexNet [87]	86.93%			
	EfficientNet-B7 [87]	93.16%			
	NASNet [87]	92.60%			
	ConvNet [87]	92.90%			
PlantVillage [16]	Multiple	Precision	Recall	F1-Score	Accuracy
	DenseNet121 [89]	99.66%	99.66%	99.66%	99.66%
	VGG16 [89]	97.99%	97.84%	97.83%	97.84%
	Xception [89]	99.28%	99.01%	98.97%	99.01%
	EfcientNet-B0 [89]	99.70%	99.69%	99.69%	99.69%
	InceptionV3 [89]	99.19%	99.17%	99.18%	99.17%
	MobileNetV2 [89]	99.27%	99.25%	99.25%	99.25%
	NASNetMobile [89]	99.46%	99.45%	99.45%	99.45%
PlantVillage [16] (Tomato only)	Multiple	Accuracy	F1-Score	Precision	Recall
	InceptionV3 [71]	93.33%	93.29%	93.38%	93.35%
	InceptionResNetV2 [71]	96.67%	96.62%	96.73%	96.67%
	MobileNetV2 [71]	91.56%	91.25%	93.25%	91.55%
	DenseNet201 [71]	85.89%	85.69%	86.10%	85.88%
PlantVillage [16] (pepper, potato, and tomato)	Multiple	Accuracy	Precision	Recall	
	EfficientNet-B4 [90]	84.03%	82.74%	84.03%	
	MobileNetV2 [90]	86.38%	87.87%	79.97%	
	ResNet50 [90]	89.02%	89.54%	83.09%	
	ResNet152V2 [90]	91.47%	90.84%	90.04%	
	DenseNet121 [90]	92.36%	93.16%	90.25%	
	MobileNet [90]	93.64%	93.79%	91.19%	
	AlexNet [90]	93.83%	93.14%	92.95%	
	DenseNet169 [90]	94.44%	93.26%	93.36%	
	PlantVillage [16]	Multiple	Accuracy		
VGG16 [92]		93.5%			
Inception [92]		94.2%			
ResNet [92]		95.1%			
EfficientNetV2 [92]		95.2%			
Proposed Approach [92]		96.4%			

achieved the second higher scores across 78 models with an average accuracy of 91.59%. Hybrid models came in at 87.83%, just under the overall average across all data with 89.80%. The lowest scores were achieved by field data with 85.22%. This showcases that lab data, with PlantVillage in particular, is easier for CNN TL models to handle, and that most every model can achieve very high results on this data, making it harder to actually compare them. Instead field and hybrid data are more difficult to learn, more close to real conditions and also offer more room for improvement, to actually compare models.

C. PRE-PROCESSING

When looking at the data pre-processing (see Table 24) that is most often applied to the datasets before training, we can see that most often, no pre-processing is used. However, among the different techniques, normalization, the process of adjusting all pixel values to be between 0 and 1, is the most common. It is said to be more optimal for DL models to handle.

In terms of train-validation-test splitting, the process of splitting data for training and validation sets, by far the most common split is 80% for training and 20% for validation. However, one can also see (Table 25) that most approaches are within a similar range.

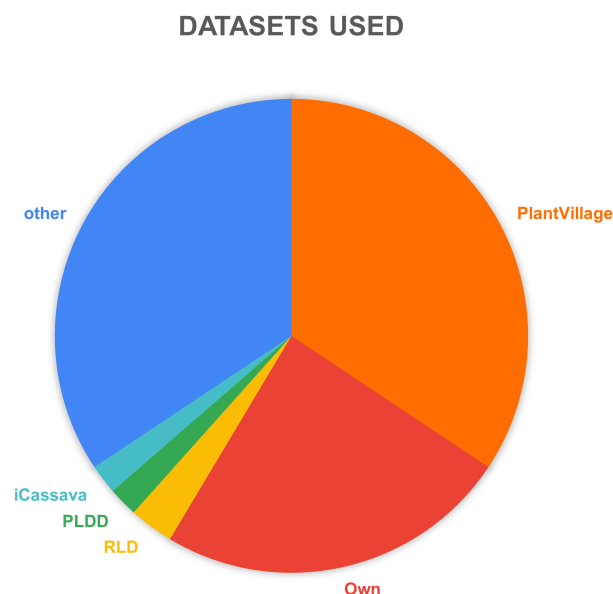


FIGURE 11. The ratio of different datasets used.

TABLE 12. The different models used in the reviewed papers and the number of times they were used. The category "other" includes all models that were used only once each.

Model	Number of times used
VGG16	34
InceptionV3	31
MobileNetV2	24
ResNet50	23
XceptionNet	20
DenseNet201	19
VGG19	19
MobileNet	15
AlexNet	13
EfficientNet-B0	13
NasNetMobile	13
InceptionResNetV2	12
DenseNet121	12
ResNet101	10
GoogleNet	6
DenseNet169	6
ResNet152V2	6
ShuffleNet	5
DenseNet	4
MobileNetV3	4
EfficientNet-B1	4
ResNet50V2	4
ResNet	3
ResNet152	3
NasNetLarge	3
EfficientNet-B2	3
EfficientNet-B3	3
EfficientNet-B5	3
EfficientNet-B7	3
EfficientNet-B6	2
EfficientNet-B4	2
EfficientNet	2
EfficientNetV2	2
MobileNetV3-Small	2
SqueezeNet	2
LeNet	2
DarkNet53	2
ResNet18	2
ResNet34	2
ResNet101V2	2
other	18

TABLE 13. The different model families used in the reviewed papers and the number of times they were used.

Model family	Number of times used
ResNet	72
Inception	71
VGG	53
MobileNet	45
DenseNet	41
EfficientNet	40
NASNet	16
AlexNet	13
ConvNeXt	2

TABLE 14. Average ranking of TL model performance on field datasets. Only models that were used at least 3 times were considered. 1st, 2nd, 3rd, and 4th refer to the 1st quartile (top 25% performance in the respective paper), 2nd quartile (top 50%), 3rd quartile (top 75%), or the 4th quartile. Num. Field is the number that model was used with field data and Avg. Field is the average quartile ranking. Quartiles were used since each study uses different models, different numbers of models, different setups, etc., so using quartiles provides some fairer comparability.

	1st	2nd	3rd	4th	Num. Field	Avg. Field
Xception	5	1	0	0	6	1.167
DenseNet169	2	1	0	0	3	1.333
EfficientNet-B0	1	2	0	0	3	1.667
DenseNet121	3	0	2	0	5	1.8
ResNet50V2	1	1	1	0	3	2
DenseNet201	2	2	2	0	6	2
MobileNetV2	1	2	2	0	5	2.2
VGG16	3	0	2	2	7	2.429
NASNetMobile	0	2	0	1	3	2.667
ResNet152V2	0	1	2	1	4	3
MobileNet	0	1	2	1	4	3
ResNet101	1	0	0	2	3	3
InceptionV3	0	2	3	3	8	3.125
ResNet50	1	1	1	4	7	3.143
VGG19	0	1	2	5	8	3.5
InceptionResNetV2	0	1	0	6	7	3.714

TABLE 15. Average ranking of TL model performance on lab datasets (excluding PlantVillage). Only models that were used at least 3 times were considered. 1st, 2nd, 3rd, and 4th refer to the 1st quartile (top 25% performance in the respective paper), 2nd quartile (top 50%), 3rd quartile (top 75%) , or the 4th quartile. Num. Lab is the number that model was used with lab data and Avg. Lab is the average quartile ranking. Quartiles were used since each study uses different models, different numbers of models, different setups, etc., so using quartiles provides some fairer comparability.

	1st	2nd	3rd	4th	Num. Lab	Avg. Lab
DenseNet201	4	0	0	1	5	1.6
EfficientNet-B0	1	1	1	0	3	2
InceptionV3	2	2	1	1	6	2.167
Xception	2	0	3	0	5	2.2
MobileNetV2	1	1	0	2	4	2.75
ShuffleNet	0	1	1	1	3	3
ResNet50	0	1	2	1	4	3
VGG16	1	1	1	3	6	3
SqueezeNet	0	1	0	2	3	3.333
NASNetMobile	0	0	1	3	4	3.75

TABLE 16. Average ranking of TL model performance on the PlantVillage dataset. Only models that were used at least 3 times were considered. 1st, 2nd, 3rd, and 4th refer to the 1st quartile (top 25% performance in the respective paper), 2nd quartile (top 50%), 3rd quartile (top 75%) , or the 4th quartile. Num. PV is the number that model was used with PlantVillage data and Avg. PV is the average quartile ranking. Quartiles were used since each study uses different models, different numbers of models, different setups, etc., so using quartiles provides some fairer comparability.

	1st	2nd	3rd	4th	Num. PV	Avg. PV
DenseNet121	4	1	0	0	5	1.2
EfficientNet-B0	3	0	1	0	4	1.5
MobileNet	2	2	0	0	4	1.5
DenseNet201	3	0	0	1	4	1.75
InceptionV3	2	3	1	1	7	2.143
ResNet50	0	3	2	2	7	2.857
MobileNetV2	1	0	2	2	5	3
Xception	1	0	0	2	3	3
VGG16	1	1	2	4	8	3.125

TABLE 17. Average ranking of TL model performance on hybrid datasets. Only models that were used at least 3 times were considered. 1st, 2nd, 3rd, and 4th refer to the 1st quartile (top 25% performance in the respective paper), 2nd quartile (top 50%), 3rd quartile (top 75%) , or the 4th quartile. Num. Hy. is the number that model was used with hybrid data and Avg. Hy. is the average quartile ranking. Quartiles were used since each study uses different models, different numbers of models, different setups, etc., so using quartiles provides some fairer comparability.

	1st	2nd	3rd	4th	Num. Hy.	Avg. Hy.
MobileNet	2	1	0	0	3	1.333
InceptionV3	2	4	0	0	6	1.667
Xception	2	0	2	0	4	2
MobileNetV2	2	2	1	1	6	2.167
VGG16	2	0	1	1	4	2.25
VGG19	0	1	1	1	3	3
InceptionResNetV2	0	1	2	1	4	3
ResNet50	0	0	2	2	4	3.5
ResNet101	0	1	0	3	4	3.5

TABLE 18. Average ranking of TL model performance on all datasets. Only models that were used at least 3 times were considered. Num. Overall is the number that model was used with all data and Avg. Overall is the average quartile overall ranking. Quartiles were used since each study uses different models, different numbers of models, different setups, etc., so using quartiles provides some fairer comparability.

	Num. Overall	Avg. Overall
EfficientNetV2	3	1
EfficientNet-B5	3	1.333
EfficientNet-B7	3	1.333
DenseNet169	7	1.429
DenseNet121	12	1.5
EfficientNet-B0	10	1.7
DenseNet201	17	1.824
Xception	18	1.944
ResNet50V2	4	2
ResNet101V2	3	2
MobileNet	13	2.308
InceptionV3	27	2.333
ResNet	3	2.333
MobileNetV2	20	2.5
EfficientNet-B4	3	2.667
VGG16	25	2.76
ResNet101	10	2.8
ResNet152V2	7	2.857
EfficientNet-B2	3	3
ResNet50	22	3.091
NASNetMobile	9	3.111
InceptionResNetV2	14	3.143
VGG19	15	3.2
ShuffleNet	4	3.25
GoogleNet	4	3.25
NASNetLarge	3	3.333
SqueezeNet	3	3.333
AlexNet	8	3.375

TABLE 19. The different datasets used in the reviewed papers and the number of times they were used. Other here stands for datasets that were not used more than once or datasets that were not specified in their papers.

Dataset	Number of times used
PlantVillage	34
Own	24
RLD	3
PLDD	2
iCassava	2
other	34

TABLE 20. The different locations that datasets used in the reviewed papers were taken in and the number of times they appeared.

Dataset location	Number of times used
India	9
China	5
Bangladesh	3
USA	3
Ethiopia	2
Turkey	2
Indonesia	2
Tunisia	1
Pakistan	1
Philippine	1
Taiwan	1
Brazil	1
Italy	1
Uganda	1
Multiple	1
Kenya	1

TABLE 21. The count of different types of datasets used in the reviewed papers. The values shown in parentheses showcase the number among "Own" datasets.

Dataset type	Number of times used
Field	31 (15)
Lab	12 (6)
Hybrid	7 (3)

TABLE 22. Average accuracy for different dataset types across papers that use at least 4 TL models for comparison and provide accuracy scores..

Dataset type	Models Applied to Dataset Type	Average Accuracy
Field	86	85.22%
Lab	78	91.59%
Hybrid	51	87.83%
PlantVillage	77	94.41%
Overall	292	89.80%

Image sizes (Table 26) are also mostly within a range that is not too far deviated from the most commonly used 224x224. 224x224, which is by far the most used size, is used so much because TensorFlow and PyTorch implementations of the pre-trained models use 224x224 pixels. With both of these libraries being the most commonly used DL frameworks, using 224x224 became somewhat of a standard. With this in mind, new datasets should ideally be at least 224x224 in size, so that one can resize them to that size easily.

D. AUGMENTATION

DL requires as much data as possible. In the field of plant disease classification, datasets are not as available as would be ideal [8], [9]. While TL does reduce the required amount of images needed, more images still train better models, and even with TL models can overfit if the data is insufficient. As such, data augmentation techniques can be used to increase the size of the datasets artificially through different methods. The resulting larger dataset can generate more robust models, without having to manually gather more data. Table 27 showcases the different methods used in the field and one can see that transformation based methods are at the

TABLE 23. Comparing the number of "Own" datasets that are shared to the public vs. ones that are kept private.

Dataset type	Number of times used
Open	4
Private	20

TABLE 24. The count of the different data pre-processing steps used in the surveyed papers. The category "other" includes all models that were used only once each.

Data pre-processing method	Number of times used
normalization	12
de-noise	3
background subtraction	3
other	7

TABLE 25. The count of the different data train-validation-test splits ratios used. The category "other" includes all models that were used only once each.

train-validation-test splits	Number of times used
80-20-0	11
70-20-10	4
60-20-20	3
80-10-10	3
90-10-0	2
70-10-20	2
other	9

TABLE 26. The number of different image sizes used. The category "other" includes all models that were used only once each.

image size	Number of times used
224x224	41
256x256	9
227x227	4
128x128	2
299x299	2
other	11

top. This includes rotation, flipping, shearing, zooming and scaling. Color based methods also see a lot of usage, with augmentations such as brightness adjustments, contrast shifting, saturation adjustments, and hue shifting also seeing use. The method of adding noise to the image has also seen a fair amount of attention in recent works. Generative Adversarial Network (GAN) which can generate new synthetic data after being trained on relevant data have also seen application, but they too require data to train the GAN, and are also much harder to implement and use than the traditional methods mentioned before, which is probably why the usage is lower. Lastly, Mix-based methods have also been explored, but much less so than transformation, color and noise based approaches.

E. METRICS

While reviewing the selected studies, we analyzed the evaluation metrics used for plant disease classification. Our find-

TABLE 27. The different data augmentation techniques used and the number of times they were used. The category "other" includes all models that were used only once each.

Augmentation technique	Number of times used
rotation	37
reflection/flip/mirror	32
shifts/translations/shear/skew	28
trim/zoom/crop	23
brightness	15
scaling	11
contrast	9
color	8
saturation	6
noise	5
transform	5
blur	3
hue	3
GAN	3
occlusion/mix based	2
other	6

TABLE 28. The different evaluation metrics used and the number of times they were used. The category "other" includes all models that were used only once each.

Evaluation metrics	Number of times used
Accuracy	80
F1-Score	77
Precision	71
Recall	64
Specificity	22
Sensitivity	11
AUC-ROC	3
Mathews Correlation Coefficient (MCC)	3
Memory Use (GPU)	2
False Positive Rate (FPR)	2
Other	13

ings show that accuracy is the most frequently used metric, followed by F1-score, indicating a strong focus on overall correctness and class balance. Many studies also incorporate precision, recall, specificity, and sensitivity to provide a more detailed performance assessment. However, relying only on accuracy could be considered insufficient in certain cases, especially for CNN and transfer learning used models where factors like class imbalance, generalization, and computational efficiency must also be considered. Less commonly used metrics such as AUC-ROC, MCC, memory usage, and FPR that can offer deeper insights into model robustness. A more comprehensive evaluation strategy, incorporating diverse metrics, is essential for reliable plant disease classification. The distribution of evaluation metrics across the reviewed papers is summarized in Table 28.

F. PLANTS

Our review, summarized in Table 29, analyzes a diverse range of plant species. We found that multi-plant species datasets are the most frequently used, appearing in 20 studies, likely due to their ability to improve model generalization but which can also partially be explained by the use of PlantVillage in most papers. Among individual plant species, rice, tomato, and potato are the most commonly studied, reflecting their importance and vulnerability to various diseases. We also observed that other frequently analyzed crops include corn, apple, chili, and coffee, while several species, such as papaya, and wheat, appear in only one study each. The dominance of certain crops suggests a research emphasis on widely cultivated and high-risk plants, whereas the limited focus on others highlights potential gaps in plant disease classification research. Based on our findings, we suggest that expanding studies to underrepresented crops could improve model robustness and applicability across diverse agricultural environments.

In terms of per plant or per disease performance, it seems as if models are generally well equipped to perform on most, if not all, plant species and disease types observed in this work. High F-1 scores that are very near the accuracy scores suggest minimal bias in must cases, proofing that TL can be applied to a wide range of plant types and diseases.

TABLE 29. The number of times that different plants were used in the papers. The category "other" includes all models that were used only once each.

Plant	Number of times used
Multi Plant	20
Rice	16
Tomato	12
Potato	4
Corn	4
Apple	3
Chili	3
Coffee	3
Soybean	3
Grape	3
Pepper	2
Banana	2
Mango	2
Cassava	2
Papaya	1
Cotton	1
Pear	1
Enset	1
Olive	1
Wheat	1
Mulberry	1
Palm	1
Banana	1

G. DATASET SIZES

The number of images used in plant disease classification research significantly impacts model performance and generalization. In our review, as summarized in Table 30, we analyzed the frequency of different dataset sizes and observed that the most commonly used range is 1,001 – 5,000 images, appearing in 29 studies. Since collecting and annotating data for deep learning tasks is time consuming and requires expert knowledge, creating very large datasets is not always feasible. It seems as if many researchers, however, consider this range of 1,001 to 5,000 images sufficient enough for training effective models. However, smaller datasets, particularly those with fewer than 1,000 images, are limited in their ability to generalize well across diverse environmental conditions, as they may not capture enough variations in disease patterns, lighting, or background noise, which is probably why they saw much less usage. On the other hand, we noticed that larger datasets, particularly those exceeding 50,000 images, were used in fewer studies but provided better model generalization by covering a broader range of conditions. Despite these advantages, working with such large datasets requires extensive computational resources, manual annotation efforts, and data storage capabilities, which can be challenging for many researchers, while also, as mentioned above, being intensively difficult to obtain. Based on our findings, we believe that while moderate dataset sizes are commonly used and practical, increasing dataset diversity and leveraging augmentation techniques or self-supervised learning could enhance model robustness. Expanding publicly available datasets and incorporating diverse imaging conditions could further improve real-world applicability, ensuring plant disease classification models perform effectively across different agricultural environments.

We analyzed the number of classes used in disease classi-

TABLE 30. Frequency of the number of images present in datasets used in the reviewed works.

Number of images range	Number of times used
200 - 1,000	5
1,001 - 5,000	32
5,001 - 10,000	19
10,001 - 20,000	7
20,001 - 50,000	7
50,001 - 60,000	12
60,001 - 90,000	8

TABLE 31. Frequency of the number of classes present in datasets used in the reviewed works.

Number of classes range	Number of times used
2 - 5	36
6 - 10	28
11 - 15	20
16 - 20	8
21 - 30	9
31 - 40	16

fication studies, as shown in Table 31. 33 studies use datasets with 2 to 5 classes, potentially due to the availability of smaller datasets. However, datasets with 6 to 15 classes are also widely used, appearing in 45 studies, indicating a growing focus on handling more diverse disease conditions. While studies with more than 20 classes are less common, they provide a more comprehensive presentation of real-world disease variability, enhancing model robustness. However, as the number of classes increases, classification complexity rises, requiring more data and advanced techniques to maintain accuracy. Expanding datasets with various class distributions could improve model generalization and better reflect real-world agricultural challenges.

H. HYPERPARAMETERS

Hyperparameters have large impact on the final performance of DL models and as such they should be chosen carefully. Bad hyperparameters could lead to bad performing models, even if the dataset, model architecture, and everything else are set up correctly. To help researchers in the field of plant leaf disease identification gain a intuitive understanding for hyperparameters, we analyzed the usage of common values for the hyperparameters used in DL. These values are not guaranteed to be the best case for new projects, but should give a good baseline understanding for where one could start. In terms of batch-size, the clear best practice seems to be 32 images per batch, being used in by far the most studies. When looking at the optimizer usage, we can again see a clear winner, that being the Adam optimizer. Learning rates generated two values that were used a significant amount of times, those being 0.0001, which was used the second most often, and 0.001, which was used the most. As for the loss function, unsurprisingly Cross-Entropy was used exclusively. All values can be found in Table 32.

Due to the effect of TL, DL models don't need as much time for training as they would, were they trained from scratch.

TABLE 32. Frequency of different hyperparameters used for training and their values.

Hyperparameter	Value	Number of times used
Batch-size	8	1
	10	4
	32	23
	16	6
	50	1
	60	1
	64	4
	100	3
	124	1
	128	1
	200	1
Optimizer	Adam	29
	SGD	9
	RMSProp	3
	SGDM	3
Learning Rate	AdaBelief	1
	0.00001	7
	0.0001	13
	0.0005	3
	0.0006	1
	0.0007	1
	0.0008	1
	0.001	22
	0.002	2
	0.04	1
Weight Decay	0.01	6
	0.1	1
	0.0002	1
	0.0005	1
Epsilon	1e-08	2
	1e-07	2
Loss Function	Cross Entropy	16
Dropout	0.5	6
	0.4	2
	0.25	1
	0.02	3
	0.1	2
	0.3	1
	1	1
	0.9	6
Regularization	Batch normalization	5
	L1:0.006	1
	L2: 0.004	1
Activation	L2: 0.01	3
	ReLU	12

TABLE 33. Frequency of epochs used for training. The category "other" includes all models that were used only once each.

Epochs range	Number of times used
10 - 50	45
51 - 100	31
101 - 200	5
201 - 500	3
501 - 1000	2
Early-stopping patience set	4

When looking at Table 33, we can see that this is consistent with the findings of our study, with most models being trained for 50 epochs or less, and with less than 100 epochs being the second most common approach. We can also see that the drop off post 100 epochs is significant, with barely any papers using a large amount of epochs before reaching satisfactory performance.

I. ANSWERING RESEARCH QUESTIONS

Q1: Which pre-trained models are currently most often used and best suited in terms of performance?

When looking at which models are the most frequently used (Figure 10 and Table 12), we can see VGG16, InceptionV3, MobileNetV2, and ResNet50 at the very top. However, while these models do see the most usage, when looking at the performance they deliver in Table 18 we can see that they all tend to be either in the middle of the ranked models or even towards the bottom. Better performing choices seem to be the EfficientNet models such as EfficientNet-B3, EfficientNet-B5, EfficientNet-B0 or DenseNet architectures such as DenseNet169, DenseNet201, or DenseNet121 (which have all seen less usage than the above mentioned more popular models). Xception has also seen good performance as well. As such, we would recommend future work to use, include comparison to DenseNet and EfficientNet models, or to base new models on said architectures.

Q2: Which dataset types are currently most often used and best suited in terms of performance?

PlantVillage is by far the most used dataset in plant leaf disease classification research. Roughly a third of all research papers use PlantVillage in their work, with roughly a second third of papers using a dataset that was collected as part of that work. Round about 80% of these "own" collected datasets are, however, kept private, immensely lowering the number of openly available data, which could be one leading factor to the heavy usage of PlantVillage. When looking at the metrics of PlantVillage studies obtained in third paper, we can see that PlantVillage, on average, generates constantly very high accuracy scores across all models. This makes it more difficult to truly evaluate and compare models using only the PlantVillage dataset. Hybrid or field data, on the other hand is more difficult for DL models to handle and therefore offer a better test bench environment, while also being closer to real life data and therefore training models that are more robust to different image conditions [18]–[20]. We therefore recommend, that researchers use hybrid or field data when possible (one could combine PlantVillage with other field and hybrid datasets to obtain such a hybrid set for example) and we hope that newly collected data will be shared more in the future, to enlarge the number of option available to researchers.

Q3: Which methods can be used to improve pre-trained models?

Among the studies reviewed in this paper, a number of them propose different methods of altering TL models in a way that improves their performance. Among these, adding attention mechanisms such as channel attention, spatial attention, channel block attention, etc., have produced good results. One can add these mechanisms after the pre-trained feature extractor and right before the classifier top, to offer the model with more selective features and therefore enhancing performance. Another option that has seen us-

age in the reviewed paper are ensemble models. Ensemble models combine different pre-existing models and merge their features maps before classifying. This allows the new ensemble model to make use of the advantages and strong points of the individual models used in the ensemble, while ideally overcoming the weaknesses. One could even consider combining both mechanisms and using a ensemble model with attention blocks added to it.

Q4: Which hyperparameters are currently most often used and best suited?

In terms of hyperparameters, clear trends have been observed in current published work. While this does not guarantee that these are best suited for any and all applications, these should offer as a good starting point. For batch-sizes, 32 was by far the most commonly used, offering a good balance between stable training and resource management. As for optimizers, the Adam optimizer has been the most used, offering strong results in combination with many different models and datasets. 0.001 was the most used learning-rate with 0.001 coming in second, offering fast enough, yet stable training. The epoch usage shows that most models managed to converge in under 50 epochs, with 51 to 100 epochs coming in second. This suggests that training models for under 100 epochs should be sufficient to train models to their full potential without overfitting. Lastly, cross-entropy is the most used loss functions, as it is designed specifically for classification tasks and therefore excel at it.

Q5: Which metrics are currently most often used and best suited?

The metrics one should consider when conduction research for TL CNN plant leaf disease classification are accuracy, F1-Score, precision, and recall. If one wishes to add additional metrics, sensitivity and specificity could be added too. We do however want to encourage future work to at least include accuracy, F1-Score, precision, and recall. Providing accuracy only does give some information about the models performance, but ignores the possible imbalance and bias to that imbalance, therefore not providing deep enough insight into the actual performance of the model. F1, recall, and precision offer that additional bias related information and should therefore be included in every publication.

Q6: Which pre-processing and augmentation techniques are currently most often used and best suited?

224x224x3 is the most commonly used input image size, which can be attributed to the fact that this is also the size of the ImageNet dataset and as a result of that also the standard size for pre-trained models. This should be considered when creating a new dataset, by making sure that images are at least of that size, or bigger. Train test splitting is carried out in 80-20 most often, with almost all papers using at least 60% of images for training and not more than 90%. Other than resizing to 224x224 and splitting many papers do not use much pre-processing, but the one technique that still does see some

usage is normalization, which can lead to better performance in some cases. In terms of augmentation techniques rotation, mirroring, translations, zooming, and brightness adjustments are most popular and have lead to well performing models. Mix-based augmentation has also proven powerful. Using these augmentation techniques, one can increase dataset size in cases where not enough data is available, while providing the models with new and more challenging image conditions, offering better domain adaptation.

Q7: Which Transfer-Learning strategies are currently most often used and best suited?

Almost all paper did use ImageNet pre-trained models, most likely due to their wide and easy availability. ImageNet, with its size and diversity of classes, offers a good basis for most tasks in TL settings. But it is also true that ImageNet is mostly domain unrelated to the task of plant leaf disease classification. TL and fine-tuning can bridge the domain gap, but using a dedicated domain-related dataset (such as PlantNet-300k for example) can lead to better performance of the domain-specific trained TL model. Sadly, such weights are, to the best of our knowledge, not openly available for download, but can be obtained by training the model one wants to use for TL on the desired pre-training dataset on their own. Then one can carry out all TL tasks with those new pre-trained weights. The downsides are that one will need to carry out the training, which could be time and hardware intensive, but these downsides could be outweighed by the potential increase in performance.

V. FUTURE WORK

After reviewing such a large number of papers and analyzing the methods these papers used, we were able to get deep insight into the field and also obtain a good picture of current trends and best practices. Through these findings, we are also able to identify research gaps and possible future directions, which we will present in this section. For one, as can be seen in Tables 12 and 13, as well as in Figure 10, some of the most used models are a bit older and have since been overtaken in terms of performance. While these models, such as VGG, can produce top-end results, we would suggest future project to at least compare them to more recent models, such as ConvNeXt for example, as well as the models that have proven better average results such as EfficientNet models, DenseNet models or, Xception to prove the validity of picking the model in question. Building new and dedicated models could also be a field of interest for future work, and using the findings of this paper as a baseline for such new models could prove helpful, knowing which architectures and hyperparameters suit the field well. Enhancing models, through methods like ensemble learning, attention blocks, or by using pre-trained weights other than ImageNet that are domain-related to the field of plant leaf disease classification could also lead to stronger models. Future studies could also implement more lightweight models, that could be used in real-time systems to improve the actual field application of these trained models.

Another field that could be improved on are datasets, which would arguably have even bigger impact on the field as a whole. As mentioned before, many of the newly collected and introduced datasets (see Table 23) are not shared with the public. While this might, in some instances, be attributable to outside factors, we would still encourage researchers, if possible, to share their datasets. With more open data, we would have a larger pool of data to train models on, which would allow researchers to train better and stronger models, but it would also allow researchers to build models for plants that have not been explored as much just yet (see Table 29). The provided augmentation statistic could help the creation of new datasets and provide valuable image condition adjustments to facilitate stronger domain adaptation even beyond the original image collection scope. With limited data, researchers could also employ techniques such as self-supervised learning, to lessen the reliance and need for large amounts of labeled data. One could also consider the possibility of multi-modal datasets, datasets that include weather data, soil data, and other data sources in combination with image data, to provide a stronger picture of the current conditions of the plants, in hopes that this can lead to better performance. Lastly, it would be ideal if a true benchmarking dataset in field or hybrid conditions, possibly with multiple plants and diseases, could be generated and shared with the public. While PlantVillage is a large size dataset with very high quality images, it does come with the downside of being taken in lab. This results in most models being able to achieve really high scores on the dataset, making it much harder to truly evaluate their performance in comparison to other models, and the resulting models are also less applicable to real in field usage. As such, having a field or hybrid benchmarking dataset to compliment PlantVillage would be desirable, so that researchers can benchmark their models on both and compare their own setup with the ones used in prior work.

VI. CONCLUSION

In this paper, we have reviewed a total of 84 papers out of a total of 118 papers originally considered for review using the PRISMA approach. Papers were collected from the Web of Science via the '("Plant Disease" OR "Leaf Disease") AND "CNN" AND "classification" AND "transfer-learning"' search term for the years 2022, 2023, 2024, and 2025. All of the 84 papers were summarized and presented including their methods and results, before identifying current trends and best practices by analyzing all papers for the methods that were used. Through this the most commonly used models (VGG16, InceptionV3, MobileNetV2) were identified, but we found that these models are not the models that show the best performance. The best performance was reported using EfficientNet family models, DenseNet family models, and Xception. This shows that there is a disconnect between the models used and the models that might be better suited, which we hope this paper can help to correct. Model improvement technologies such as ensemble learning, attention mechanisms, and domain-related pre-training were pointed out and

presented. The most popular datasets (especially PlantVillage) were identified, but it was also shown that lab datasets were by far the easiest to learn, with PlantVillage being even easier specifically, offering less of a challenge to models and making benchmarking studies, comparative studies, or even feasibility studies a bit less meaningful, since this data is so much more simple to handle. Field and hybrid data would be recommended to be used when possible. We also show-cases that most of the newly collected data is kept private, lowering the number of data open to the public, a trend that hopefully does not continue. This paper also presents the most common locations for data collection (India, China, USA), as well as the distribution of dataset types (Field > Lab > Hybrid). Additionally, we also ranked the popularity for pre-processing techniques (normalization, de-noise, background subtraction), train-validation-test splits (80-20-0, 70,20-10, 60-20-20), image sizes (224x224, 256x256, 227x227), and augmentation techniques (rotation, flipping, shearing). Metrics (accuracy, F1-Score, precision), plant types (multi-plant, rice, tomato), as well as hyperparameters which were also analyzed and ranked. This offers deep insight into the current state of plant leaf disease classification using TL. Using these findings, researchers can create new projects based on the best practices and iterate and improve upon them.

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